

Transitory and Persistent Cross-Market Spillovers of Energy Tokens

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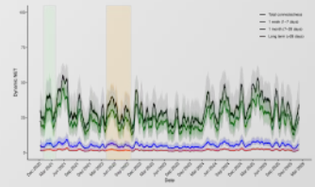
IDA Institute Digital Assets
Bucharest University of Economic Studies
MSCA Digital Finance, Masaryk University

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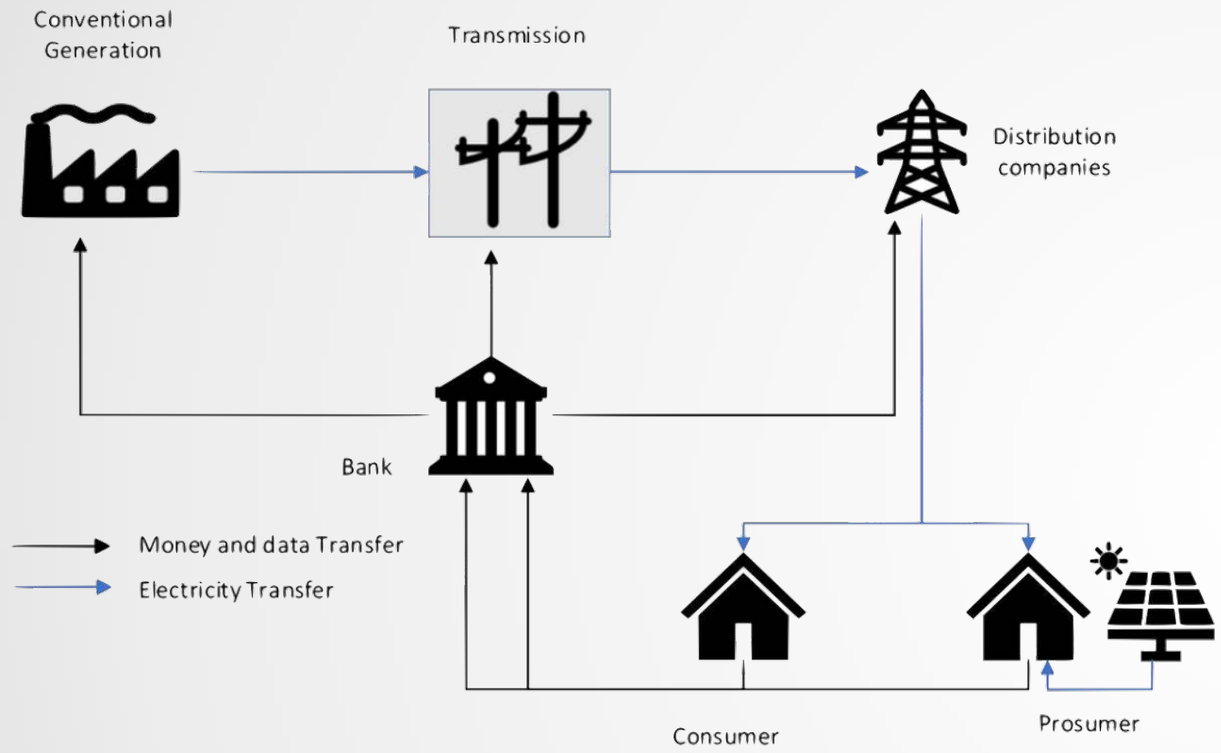
Energy Transition in a Low-Carbon World

- Renewable energy
- Challenges
  - high costs
  - difficulties in managing surplus electricity
  - long payback periods
  - make investments risky
- Blockchain technology!
  - Make renewable energy more efficient, transparent, and attractive!

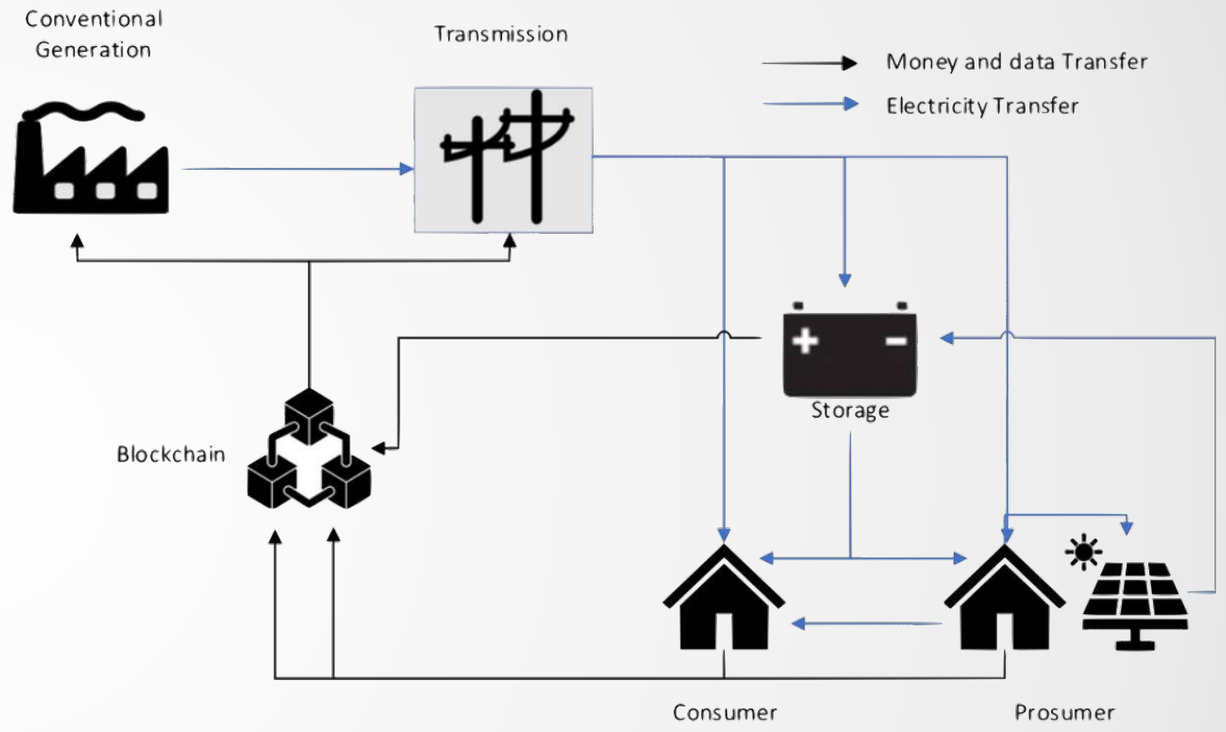


Blockchain in Energy Market

Tokenization



(a) Conventional Energy Trading

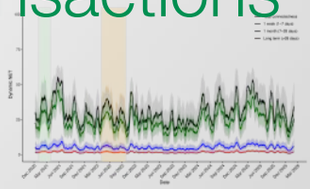


(b) Blockchain based P2P energy trading

Energy market scenario

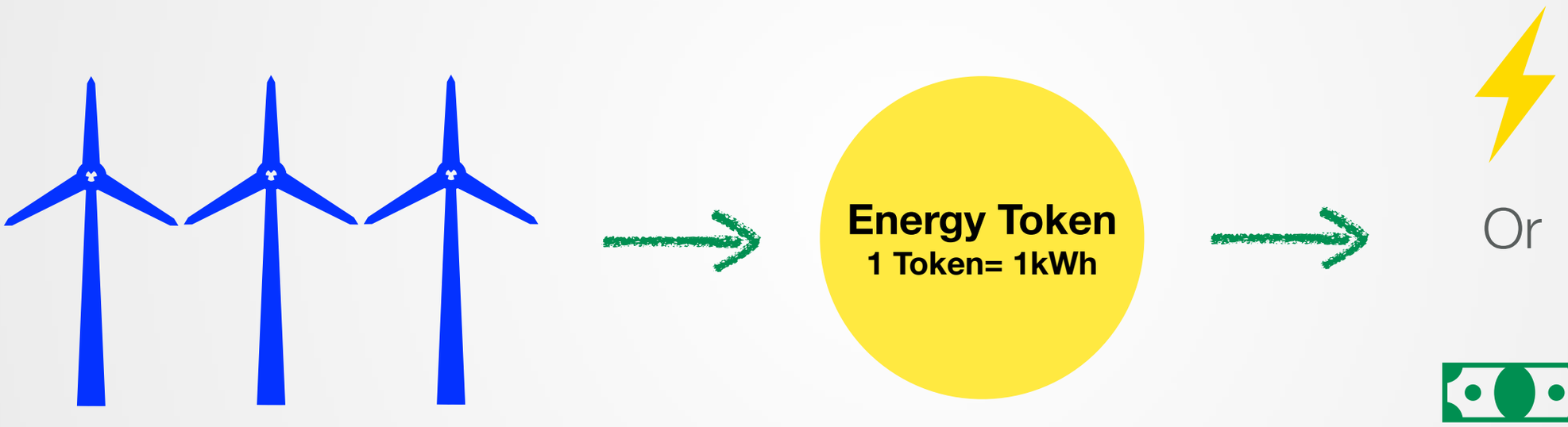
- Centralised grid & banks
- Slow, costly, no local flexibility

- Peer-to-peer trading
- Storage supports renewables
- Smart contracts ensure secure
- transparent, low-cost transactions



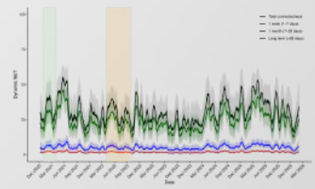
Energy Tokenization in Blockchain

- Converts renewable energy into **digital tokens**
- Each token = a self-executing power-purchase contract
- **Tradeable**: price set by supply & demand



Energy Tokens Issued

- Clear ownership of energy
- Easier & faster energy trading
- More liquidity and investment efficiency
- Wider investor access



Energy Token as Crypto

Cryptocurrencies

Transaction mechanism secure, and decentralized peer-to-peer payments.

Coins

mining coins

non-mining

Bitcoin
Litecoin

Ripple,
Binance Coin

Tokens

Tokens are digital assets created through smart contracts on a blockchain

utility tokens

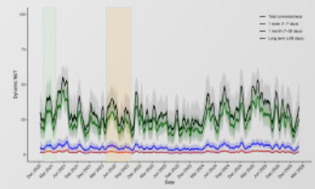
security tokens

fungible tokens

NFTs

Golem,
FileCoin
Storj

Energy Tokens



Market Interconnectedness

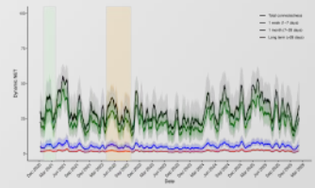
- Renewable energy tokens can provide portfolio diversification hedging tools during market downturn (Ustaoglu. 2024)

- During extreme events (COVID-19, Russia–Ukraine war).

- Return spillovers increase significantly (Yang et al. 2025)

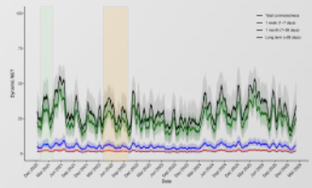
- Role of energy tokens in diversification strategies and the importance of monitoring their connectedness

- Supported by blockchain adoption in energy sector & investor focus on renewables, ESG, and green finance.



Objective

- Examine the **connectedness** among:
  - Renewable energy tokens
  - Cryptocurrencies
  - Renewable energy stocks
  - Commodities
  - ESG & green financial assets
- Relationships under different market conditions: Cryptocurrency turmoil 2021 and global energy crisis 2022.
- Capture the smoothly varying persistence structure of dynamic linkages
- Develop a frequency-domain framework to distinguish between: Transitory components and Persistent components
- Evaluate implications for portfolio diversification.



Outline

1. Motivation ✔
2. Methodology
3. Data
4. Empirical Results
5. Policy Implications
6. Conclusion.

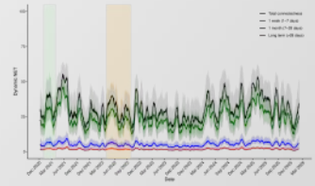
Measuring transitory and persistent connections

- $\boldsymbol{X}_{t,T}$ locally stationary TVP-VAR, Baruník & Ellington 2024

$$\boldsymbol{X}_{t,T} = \boldsymbol{\Phi}_1(t/T)\boldsymbol{X}_{t-1,T} + \cdots + \boldsymbol{\Phi}_p(t/T)\boldsymbol{X}_{t-p,T} + \varepsilon_{t,T}$$

  - t discrete time index
  - T is an additional index indicating the sharpness of the local approximation of the time series $\boldsymbol{X}_{t,T}$ by a stationary one.

- $\boldsymbol{\Phi}_1(t/T)$ and $\varepsilon_{t,T}$ are the time varying autoregressive coefficients.
- Construct frequency-dependent, time-varying measures of network connectedness.



Measuring transitory and persistent connections

- Locally stationary processes, assuming approximate stationarity within short intervals.
- In a neighbourhood of a fixed time point $u_0 = t_0/T$, we approximate the process $\boldsymbol{X}_{t,T}$ by a stationary process $\widetilde{\boldsymbol{X}}_t(u_0)$ as:

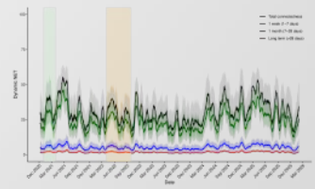
$$\widetilde{\boldsymbol{X}}_t(u_0) = \boldsymbol{\Phi}_1(u_0)\,\widetilde{\boldsymbol{X}}_{t-1}(u_0) + \cdots + \boldsymbol{\Phi}_p(u_0)\,\widetilde{\boldsymbol{X}}_{t-p}(u_0) + \boldsymbol{\varepsilon}_t,$$
$$t \in \mathbb{Z}$$

- Under suitable regulatory conditions:

$$\left|\boldsymbol{X}_{t,T} - \widetilde{\boldsymbol{X}}_t(u_0)\right| = \mathcal{O}_p\left(\left|t/T - u_0\right| + 1/T\right)$$

  - "locally stationary process"





Measuring transitory and persistent connections

The process has time varying $VMA(\infty)$ representation:

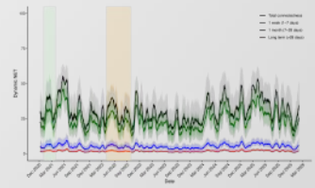
$$\boldsymbol{X}_{t,T} = \sum_{h=-\infty}^{\infty} \boldsymbol{\Psi}_{t,T}(h)\varepsilon_{t-h}$$

$\boldsymbol{\Psi}_{t,T}(h)$ is a stochastic process:

$$\sup_{\ell} \|\boldsymbol{\Psi}_t - \boldsymbol{\Psi}_\ell\|^2 = \mathcal{O}_p(h/t) \text{ for } 1 \leq h \leq t \text{ as } t \to \infty,$$

$$\boldsymbol{\Psi}_{t,T}(h) = \left\{\boldsymbol{\Phi}_{t,T}(h)\right\}^{-1}$$

key to understanding dynamics.

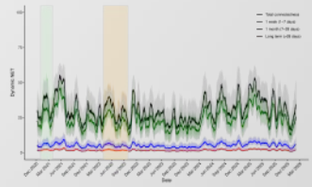


Frequency Response

- Shocks are not orthogonal → variance decomposition requires an identification scheme.
- Adapt generalized identification to locally stationary processes.
- Instead of impulse responses, use **local frequency response function**:

$$\Psi_{t,T}\left(e^{-i\omega}\right) = \sum_{h} e^{-i\omega h}\Psi_{t,T}(h)$$

- Obtained via Fourier transform of VAR coefficients.
- Frequency response separates **transitory (high-ω)** vs. **persistent (low-ω)** connections.



Time-Varying Spectral Density

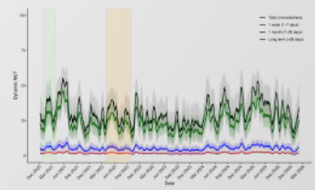
- Local spectral density of $\boldsymbol{X}_{t,T}$ at frequency ω:

$$\boldsymbol{S}_{\boldsymbol{X}}(u,\omega) = \sum_{h=-\infty}^{\infty} E\left[\widetilde{\boldsymbol{X}}_{t+h}(u)\widetilde{\boldsymbol{X}}_{t}^{\top}(u)\right] e^{-i\omega h}$$

- Equivalent form (via frequency response):

$$\boldsymbol{S}_{\boldsymbol{X}}(u,\omega) = \left\{\boldsymbol{\Psi}\left(u,e^{-i\omega}\right)\right\}\boldsymbol{\Sigma}(u)\left\{\boldsymbol{\Psi}\left(u,e^{+i\omega}\right)\right\}^{\top}$$

- Distribution of the time varying covariance of $\boldsymbol{X}_{t,T}$ over frequency components ω.
  - Building block for frequency-dependent variance decompositions
  - How much of the system's variability comes from short-run versus long-run movement?

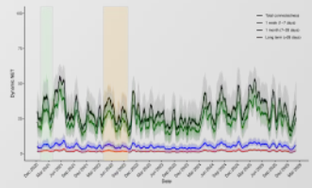


From Spectral Density to Networks

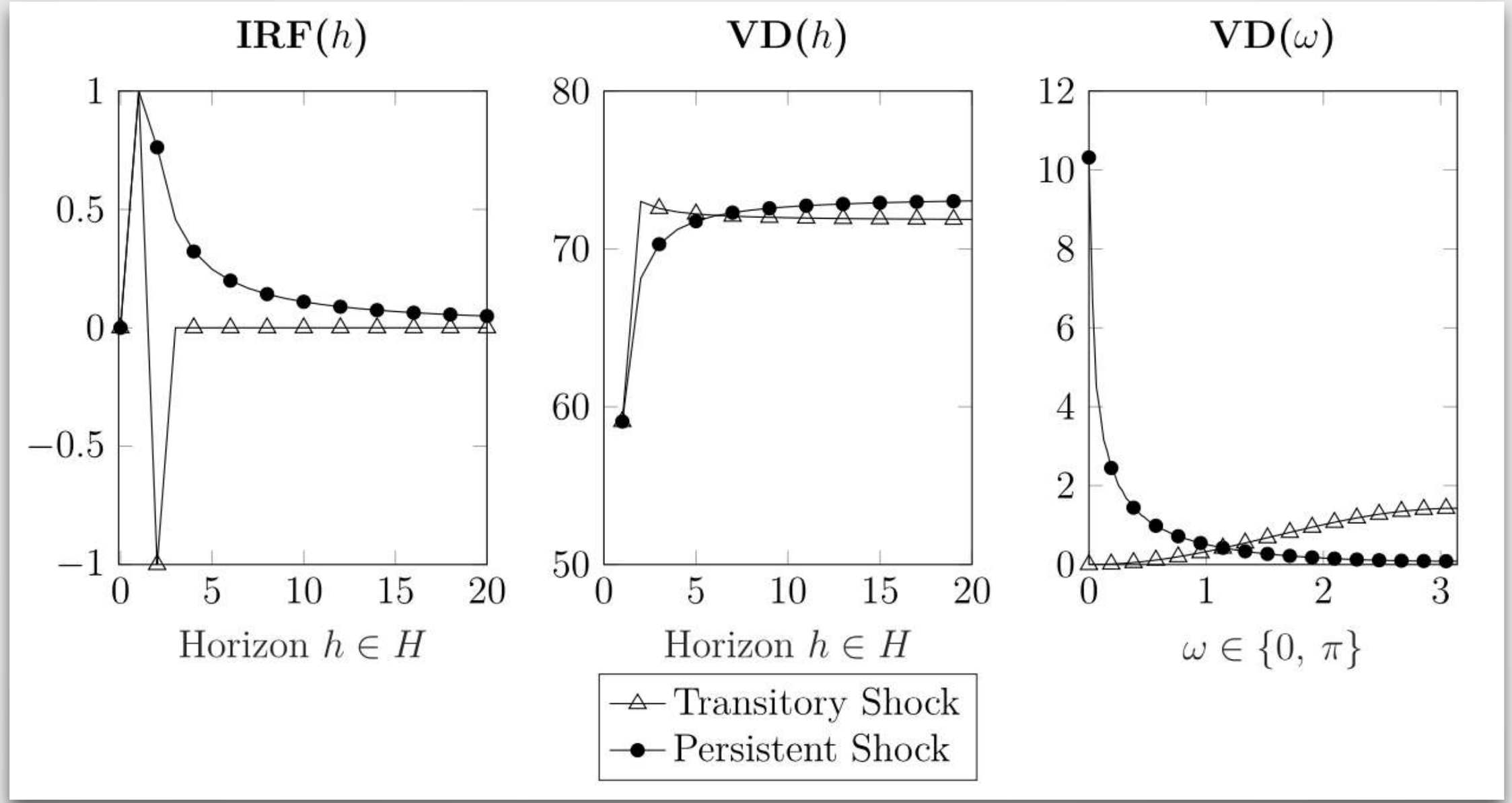
- Variance decompositions ⟡ adjacency matrices (Diebold & Yilmaz, 2014):
  - Off-diagonal elements ≻ cross-variable spillovers.
  - Diagonal elements ≻ own-shock contributions.

- In spectral domain:
  - Adjacency matrices become time- and frequency-dependent.
  - Enables dynamic networks with persistence structure:
    1. High-frequency bands ≻ transitory connectedness.
    2. Low-frequency bands ≻ persistent connectedness.

- Spectral density provides the building block for measuring systemic risk across horizons.



VD: Time vs frequency domain.



Impulse Response Functions and Variance Decompositions in time and frequency domain.

Source: Baruník & Ellington (2024)

Dynamic adjacency matrix

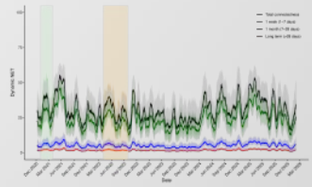
- Time-varying spectral representation of the VD of shocks from variable j to variable k at a rescaled time $u = t_0/T$ (network measures in the time–frequency domain).

impact of shocks from k on j at frequencies d=(a,b)

$$[\boldsymbol{\theta}(u,d)]_{j,k} = \frac{\sigma_{kk}^{-1} \int_a^b \left| \left[\boldsymbol{\Psi}\left(u, e^{-i\omega}\right) \boldsymbol{\Sigma}(u) \right]_{j,k} \right|^2 d\omega}{\int_{-\pi}^{\pi} \left[\left\{ \boldsymbol{\Psi}\left(u, e^{-i\omega}\right) \right\} \boldsymbol{\Sigma}(u) \left\{ \boldsymbol{\Psi}\left(u, e^{+i\omega}\right) \right\}^{\top} \right]_{j,j} d\omega}$$

Total variance j

- Links change over time and across frequencies.
- How relationships evolve during stable periods and during crises



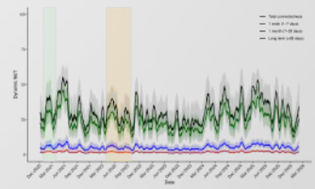
Dynamic adjacency matrix

- The adjacency matrix is **directional**: the link $j \rightarrow k \neq k \rightarrow j$: the adjacency matrix is **asymmetric.**
- If we add up connectedness across all frequency bands, we get the **total time-domain connectedness** (the Diebold–Yilmaz framework):

$$\Theta(u) = \sum_{d_s \in D} \Theta\left(u, d_s\right)$$

  - The adjacency matrix is time-varying and frequency-specific,

- Study of network characteristics at different frequency bands.
  - In the short-run, transitory network connections (High frequency).
  - In the long-run, persistent network connections (low frequency).



Connectedness measures

- Total Connectedness: overall spillovers in the system:

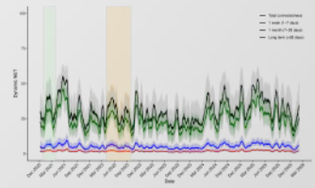
$$C(u,d) = 100 \times \frac{\sum_{j \neq k} \tilde{\Theta}_{j,k}(u,d)}{\sum_{j,k} \tilde{\Theta}_{j,k}(u)}$$

  - Higher value $\succ$ more interconnected system $\succ$ higher systemic risk.

- From-Connectedness (Receiving shocks, out-degrees):

$$C_{j \leftarrow \bullet}(u,d) = 100 \times \frac{\sum_{k \neq j} \tilde{\Theta}_{j,k}(u,d)}{\sum_{j,k} \tilde{\Theta}_{j,k}(u)}$$

  - How much of variable j's variance is explained by others.
  - Shows which assets are mainly receivers of shocks



Connectedness measures

- To-Connectedness (Transmitting shocks, in-degrees):

$$C_{j\rightarrow\bullet}(u,d) = 100 \times \frac{\sum_{k\neq j} \tilde{\Theta}_{k,j}(u,d)}{\sum_{j,k} \tilde{\Theta}_{j,k}(u)}$$

  - How much of variable j's variance is contribute to variance in others.
- Net Connectedness: j is mainly a **transmitter** or a **receiver** of shocks.

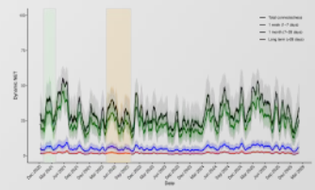
$$C_j^{net}(u,d) = C_{j\rightarrow\bullet}(u,d) - C_{j\leftarrow\bullet}(u,d)$$

- We can reconstruct time domain network connectedness measures from our frequency-dependent networks.

$$c(u) = \sum_{d_s \in D} c\left(u, d_s\right)$$

$$c_{j\leftarrow}(u) = \sum_{d_s \in D} c_{j\leftarrow\cdot}\left(u, d_s\right)$$

$$c_{j\rightarrow\cdot}(u) = \sum_{d_s \in D} c_{j\rightarrow\cdot}\left(u, d_s\right)$$



Testing for Statistical Differences in Connectedness

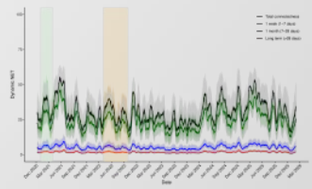
- Difference between transitory and persistent connectedness: statistically significant / just random noise.
- Null hypothesis: $H_0 : C\left(u, d_a\right) = C\left(u, d_b\right)$
- Wald-type statistic (from Bayesian posterior):

$$W = \left(\bar{D}(u) - D_0(u)\right)^{\top} \left[V_D(\bar{D}(u))\right]^{-1} \left(\bar{D}(u) - D_0(u)\right)$$

$$W \sim \chi^2_q$$

$$D(u) = C\left(u, d_a\right) - C\left(u, d_b\right)$$

- Large $W \succ$ reject H_0: connectedness is significantly different across horizons.
- Small $W \succ$ fail to reject H_0: no meaningful difference.

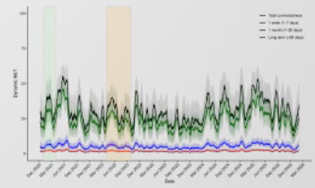


Data and Sample Period

- Assets:
  - Renewable energy tokens
  - Cryptocurrencies
  - Renewable energy stocks
  - Commodities
  - ESG & green financial assets

- Sample period:
  - 07.12.2020 to 31.01.2026

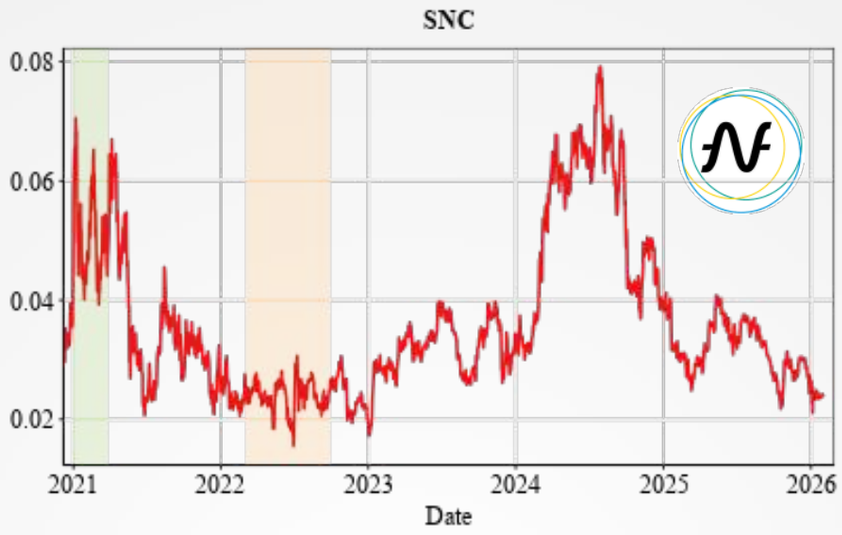
- Two Events:
  - Q1 2021: Cryptocurrency Turmoil
  - Q2-Q3 2022: Energy Crisis.



Energy Tokens



Power Ledger (POWR)



SunContract (SNC)



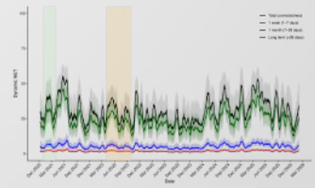
Energy Web Token (EWT)



Efforce (WOZX)



WePower (WPR)



Cryptocurrency



Bitcoin Exchange Traded Fund ETF

Bitcoin (BTC)



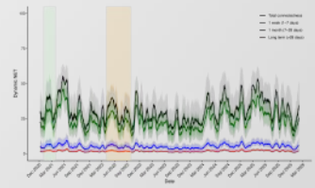
Ethereum (ETH)



Solana



Avalanche



Cryptocurrency Index



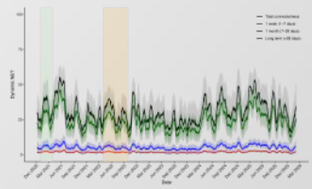
DeFi Pulse Index



IDEX



S&P Cryptocurrency Top 20 Index



Energy Stocks and Indices



First Trust Nasdaq Clean Edge Green Energy Index Fund (QCLN)



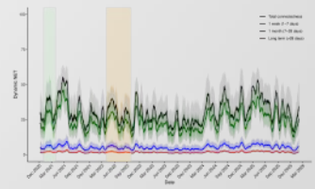
Invesco Global Clean Energy ETF (PBD)



Enphase Energy (ENPH)



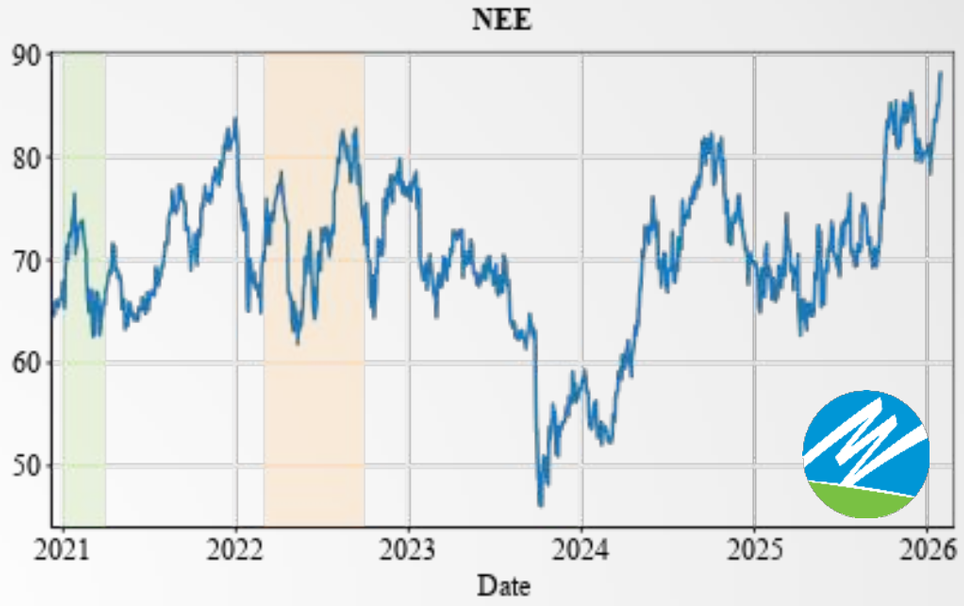
Invesco Wilderhill Clean Energy ETF (PBW)



Energy Stocks and Indices



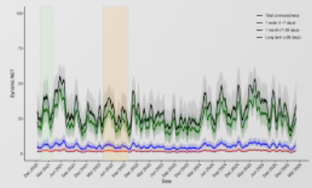
Fidelity MSCI Energy Index ETF (FENY)



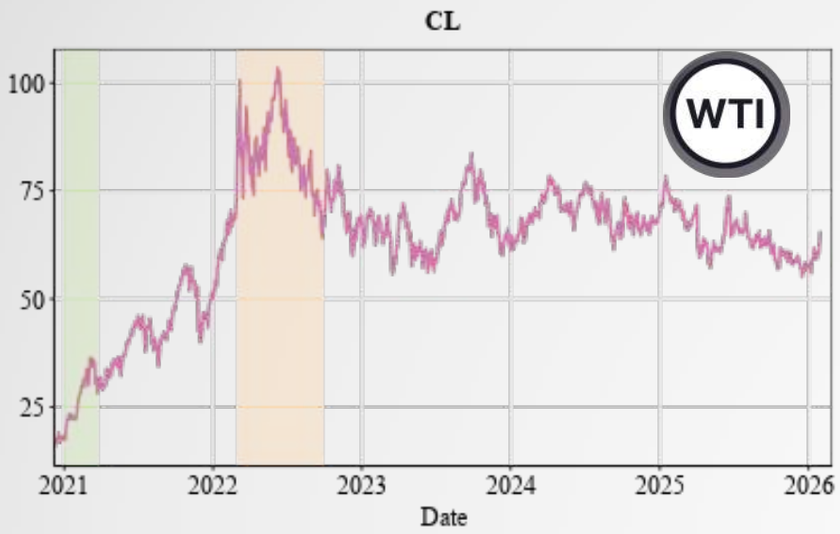
Nextera Energy (NEE)



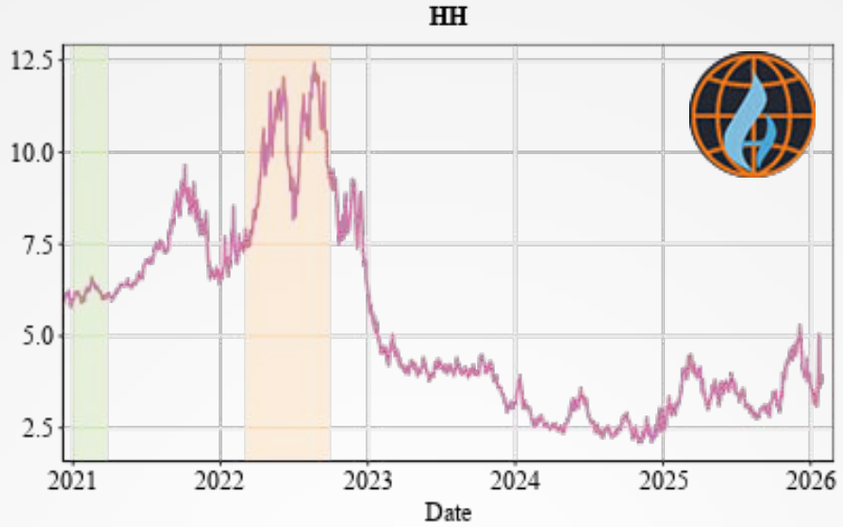
iShares Global Clean Energy ETF (ICLN)



Commodities



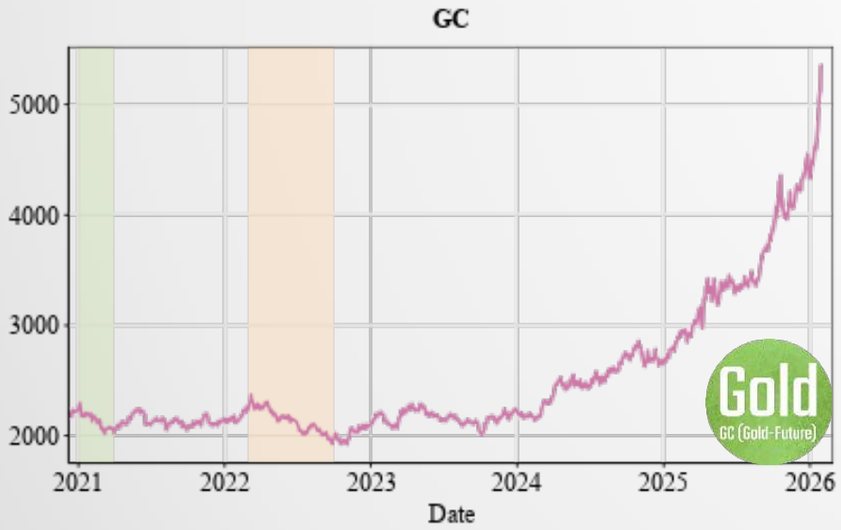
Crude Oil WTI Futures (CL)



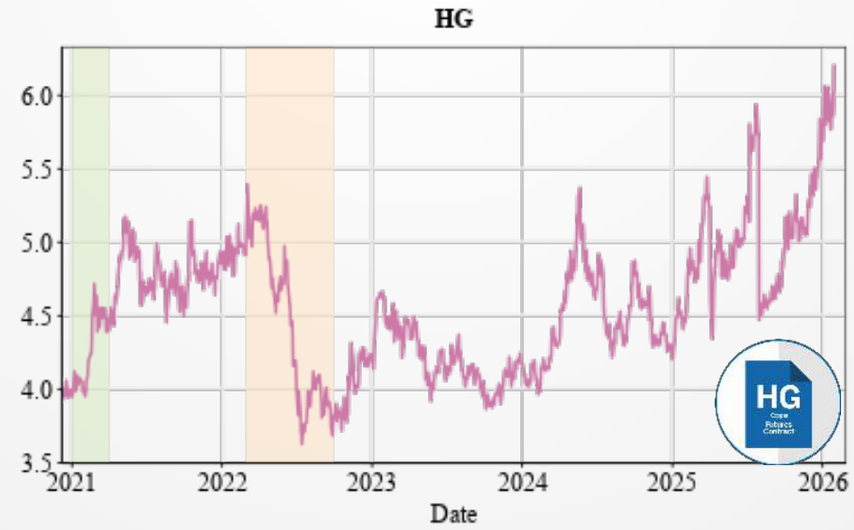
Natural Gas Henry Hub Futures (HH)



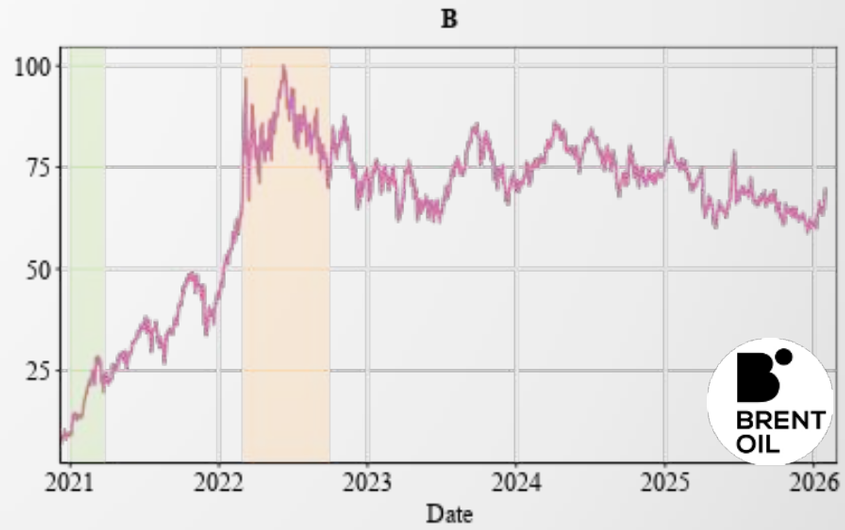
Global X Lithium ETF (LIT)



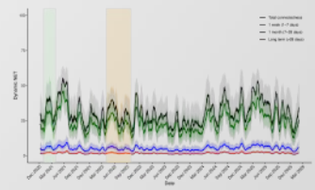
Gold Futures (GC)



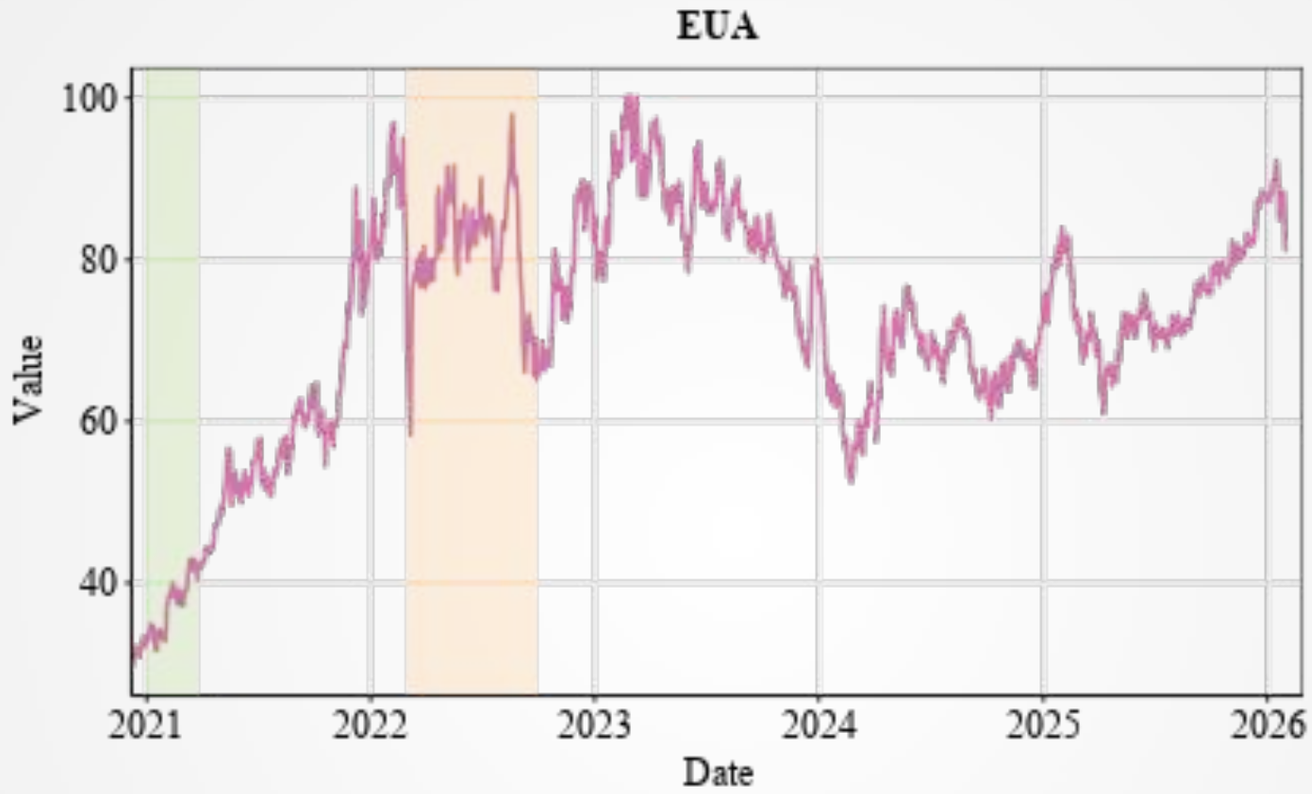
Copper Futures (HG)



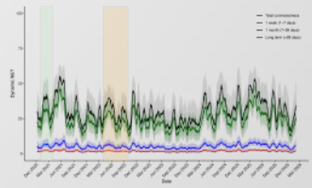
Brent Crude (B)



Commodities



European Union Allowance (EUA)



ESG and Green Finance Assets



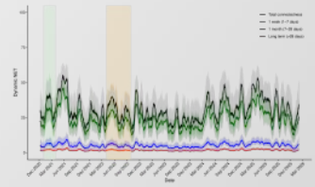
iShares Global Green Bond ETF (BGRN)



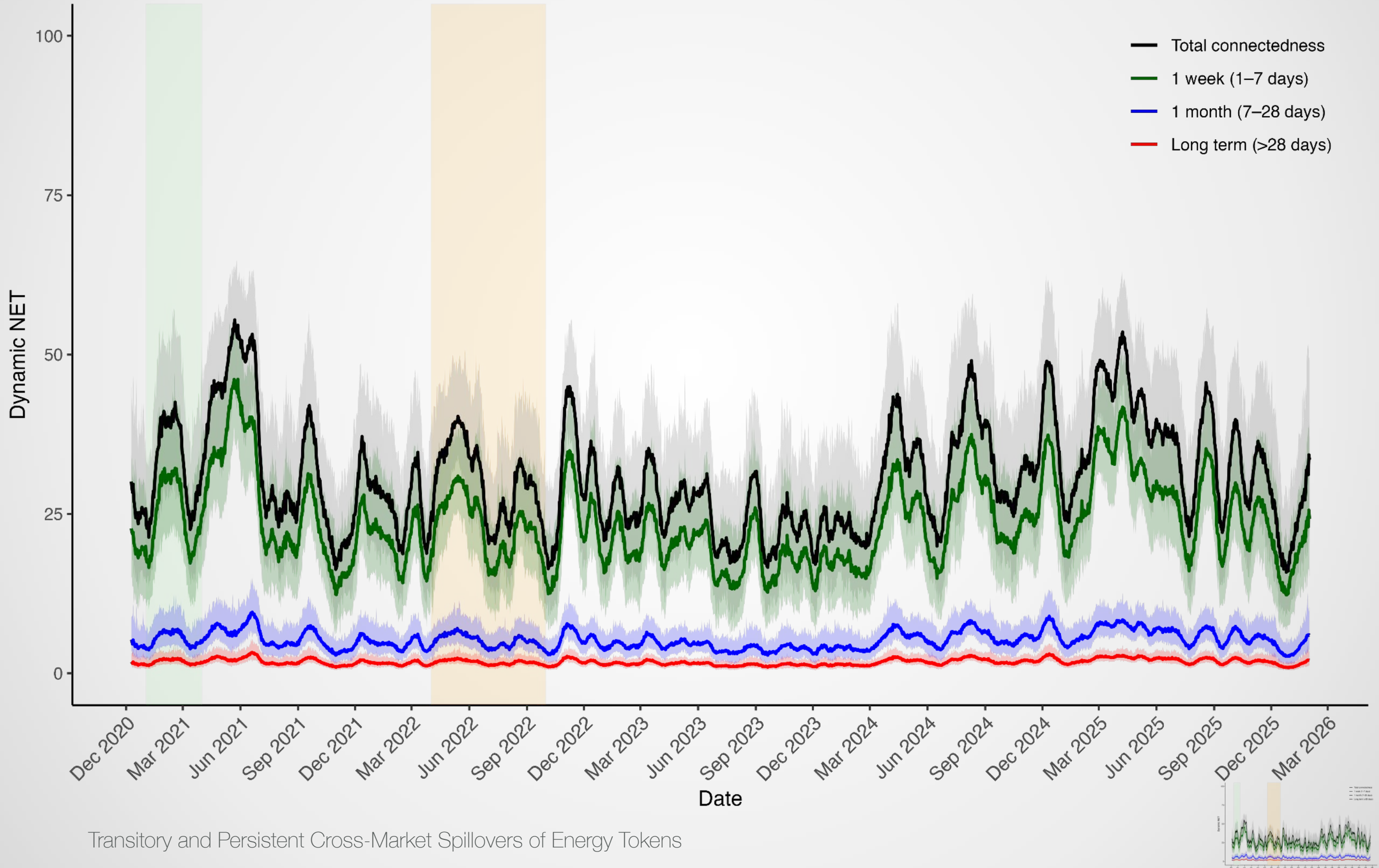
iShares Trust iShares ESG Aware MSCI Eafe ETF (ESGD)



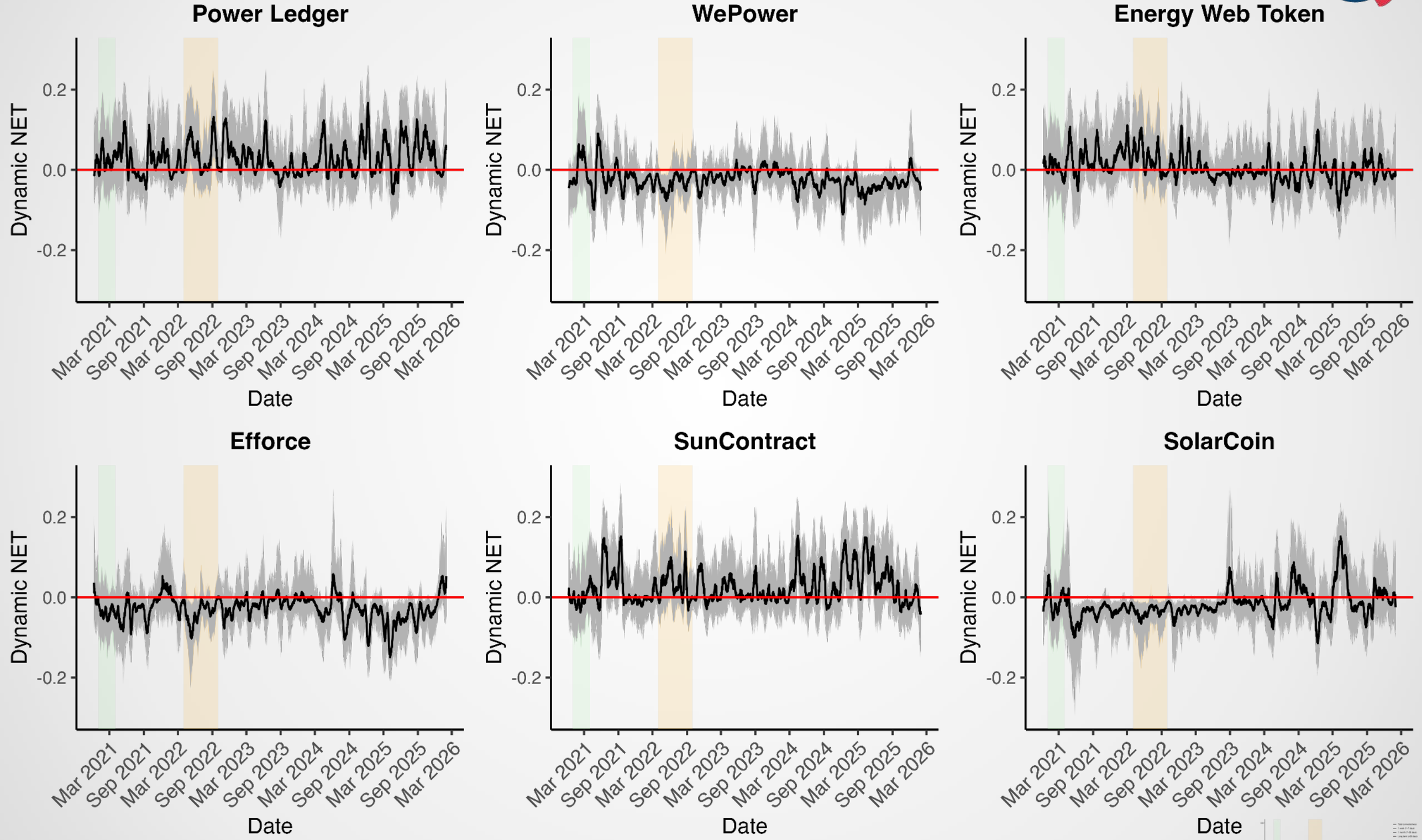
iShares ESG MSCI Usa ETF (ESGU)



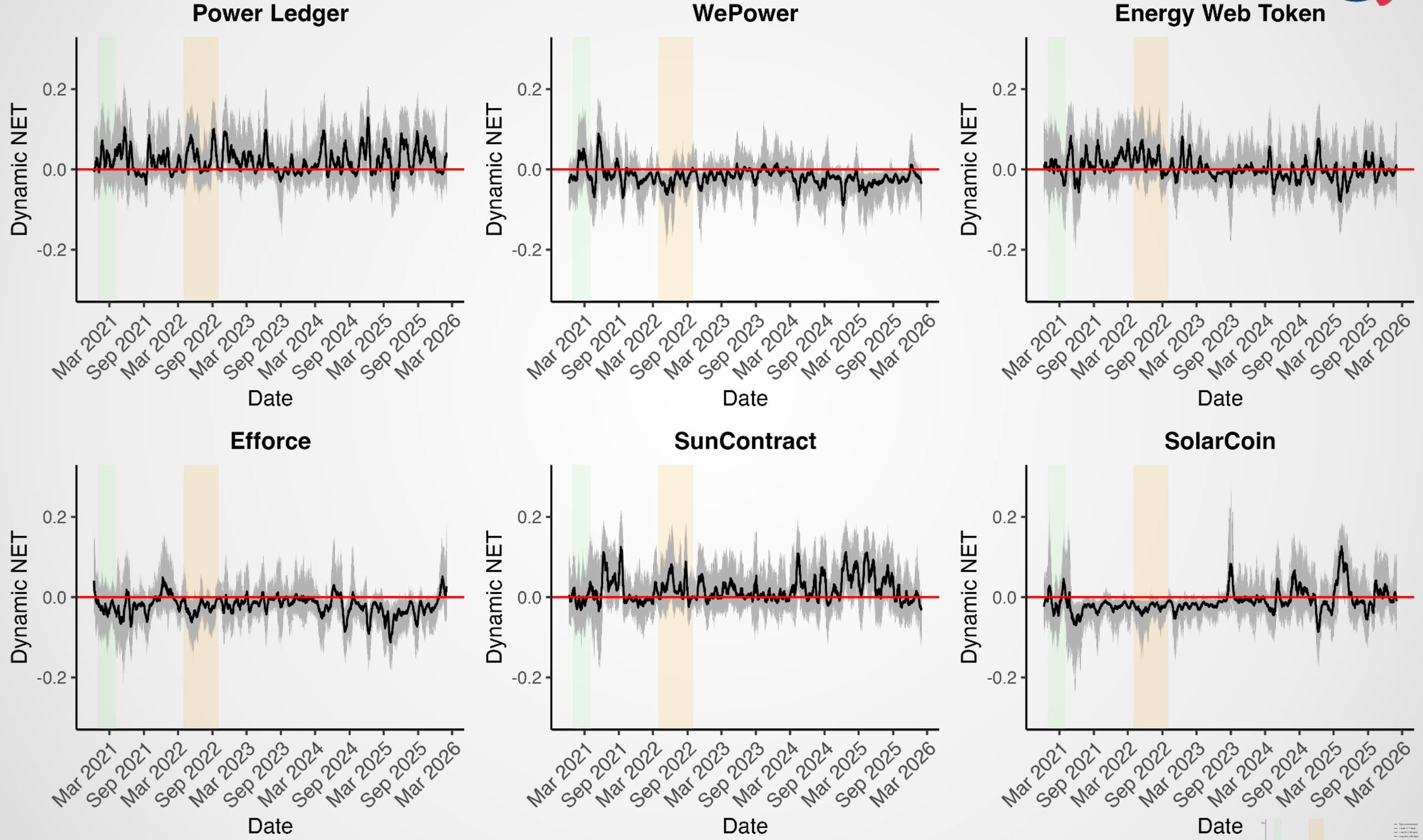
Energy Tokens - Dynamic Network Connectedness



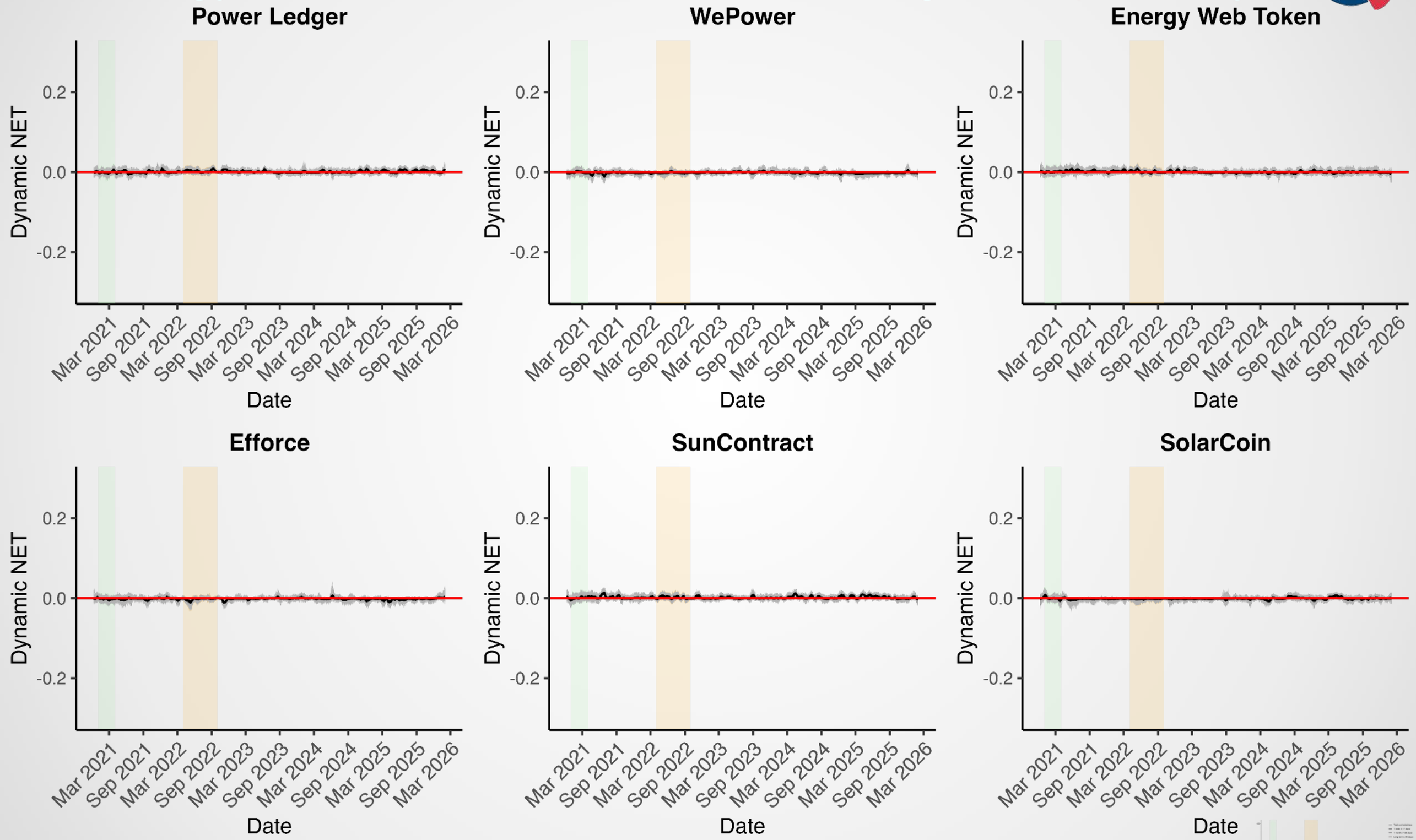
Energy Tokens - Directional NET - 1 week (1–7 days)



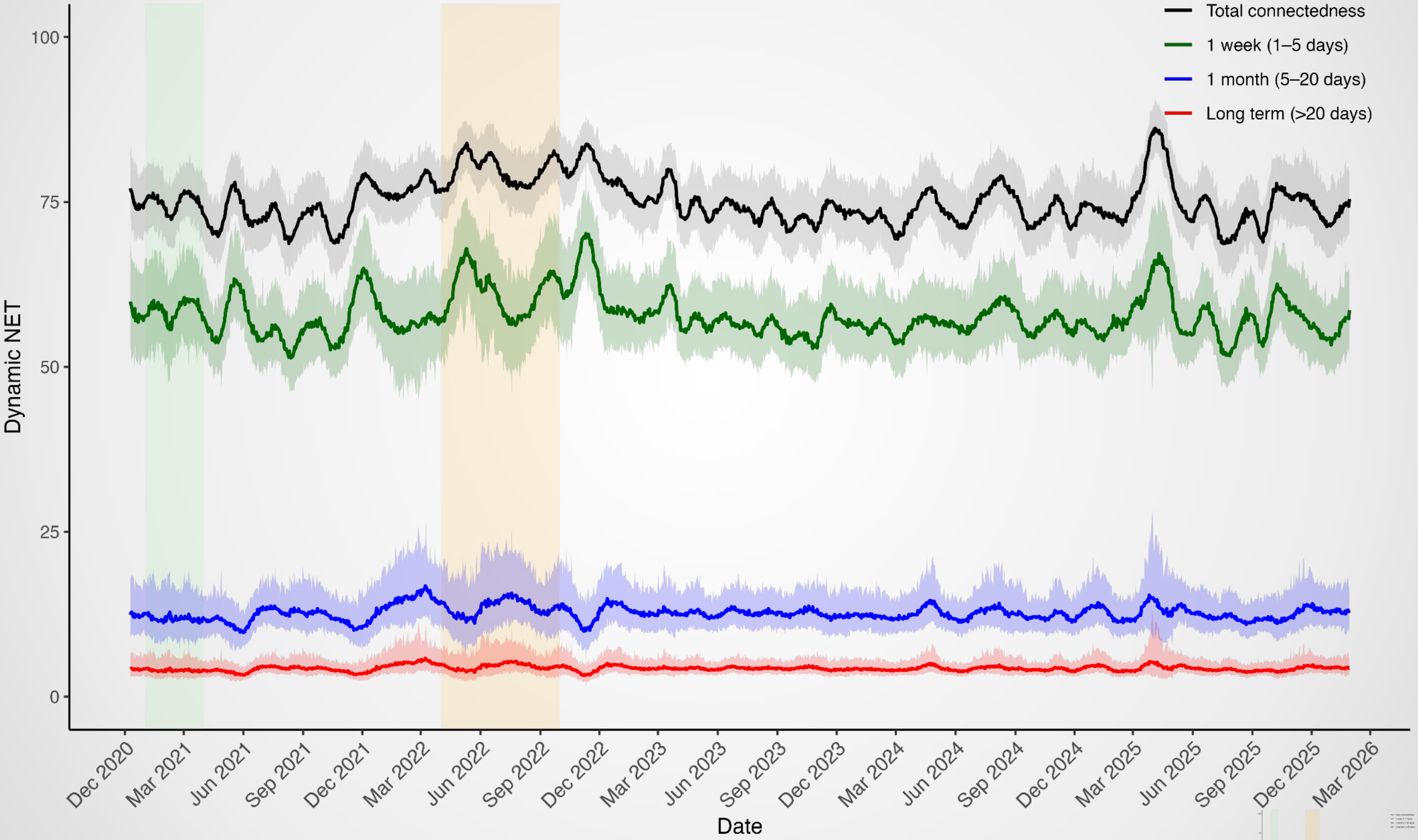
Energy Tokens - Directional NET - 1 month (7–28 days)



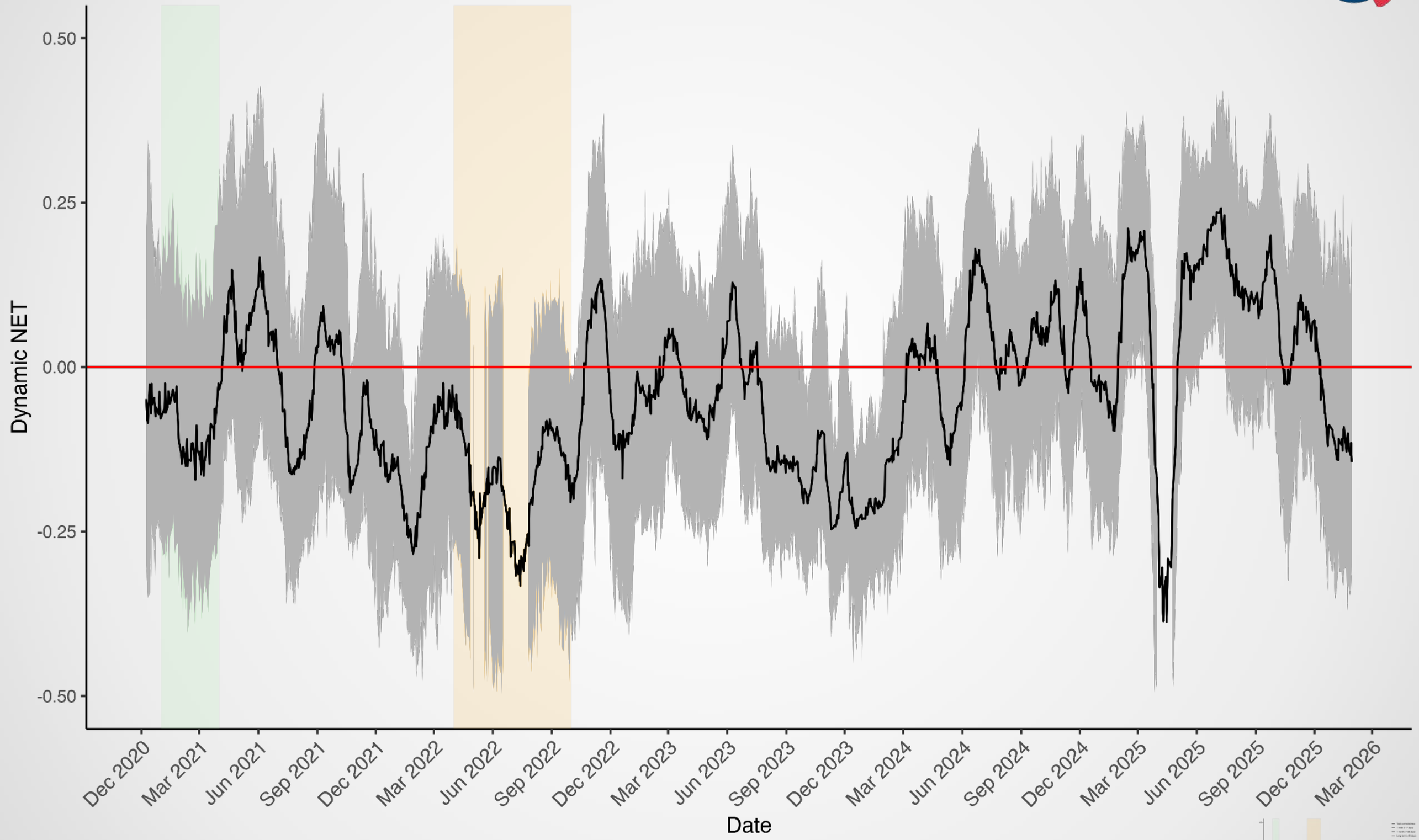
Energy Tokens - Directional NET - Long term (>28 days)



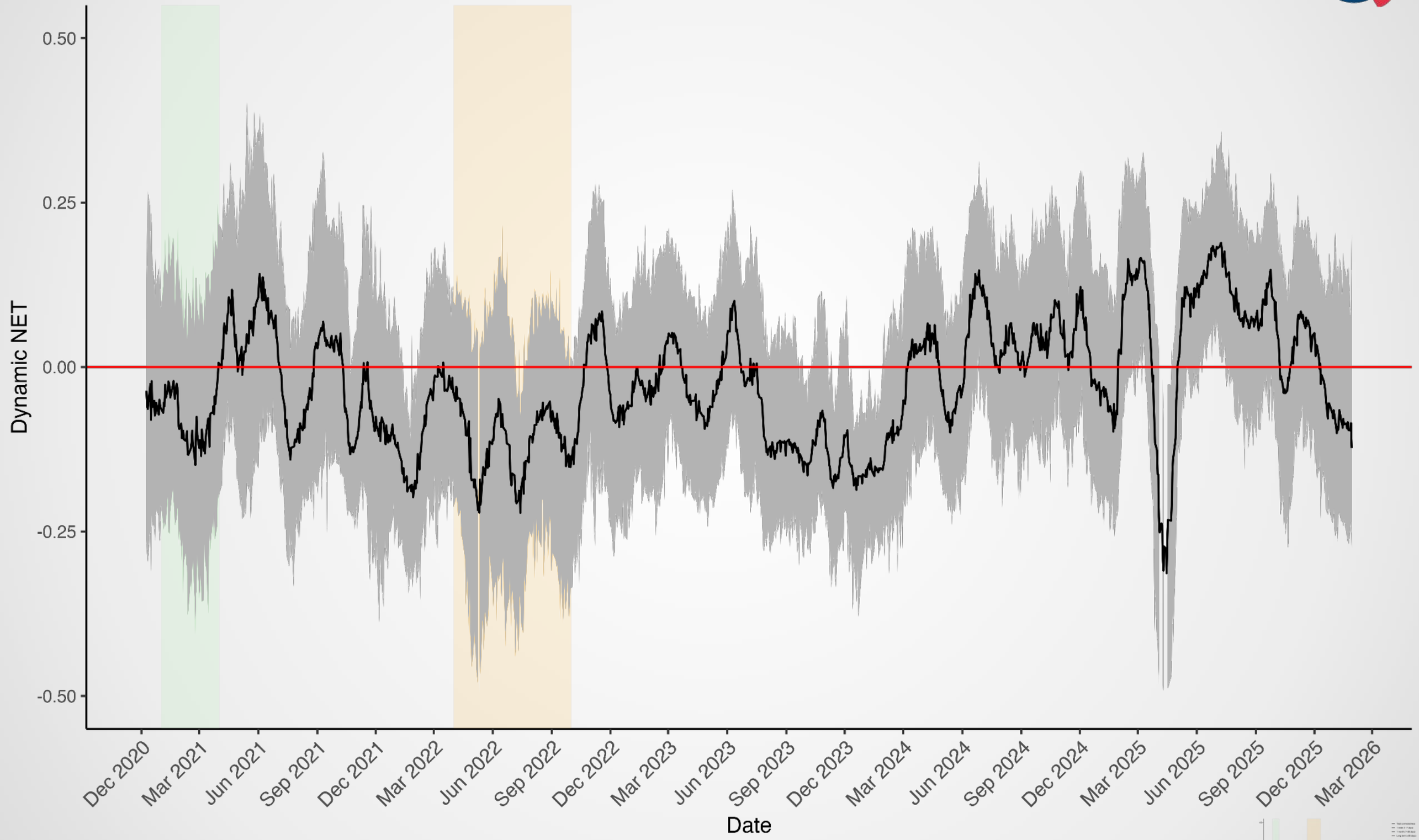
Power Ledger - Dynamic Network Connectedness



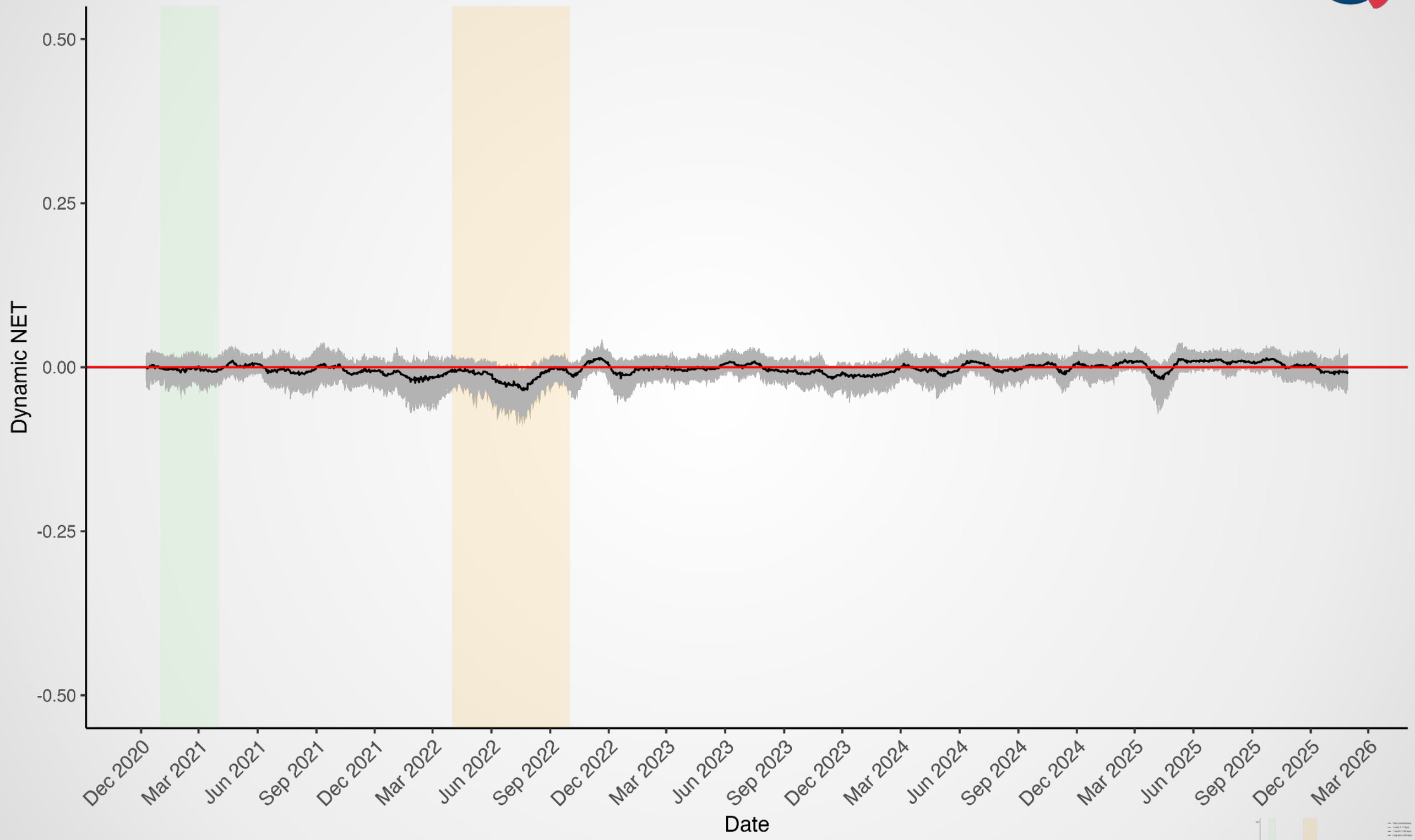
Power Ledger - Directional NET - 1 week (1–5 days)



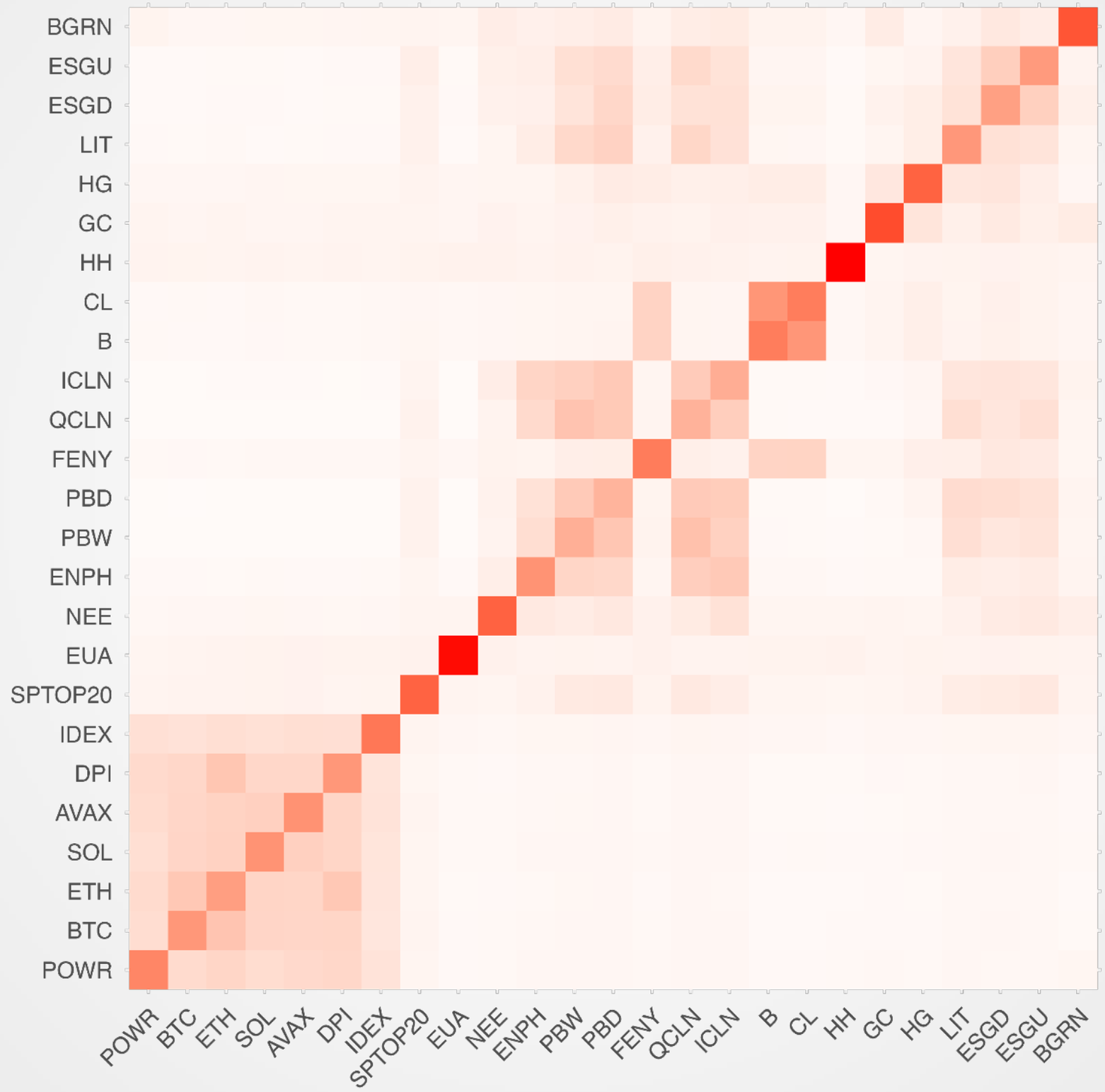
Power Ledger - Directional NET - 1 month (5–20 days)



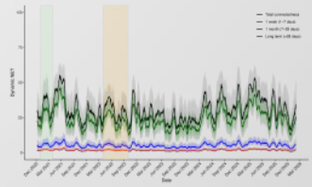
Power Ledger - Directional NET - Long term (>20 days)



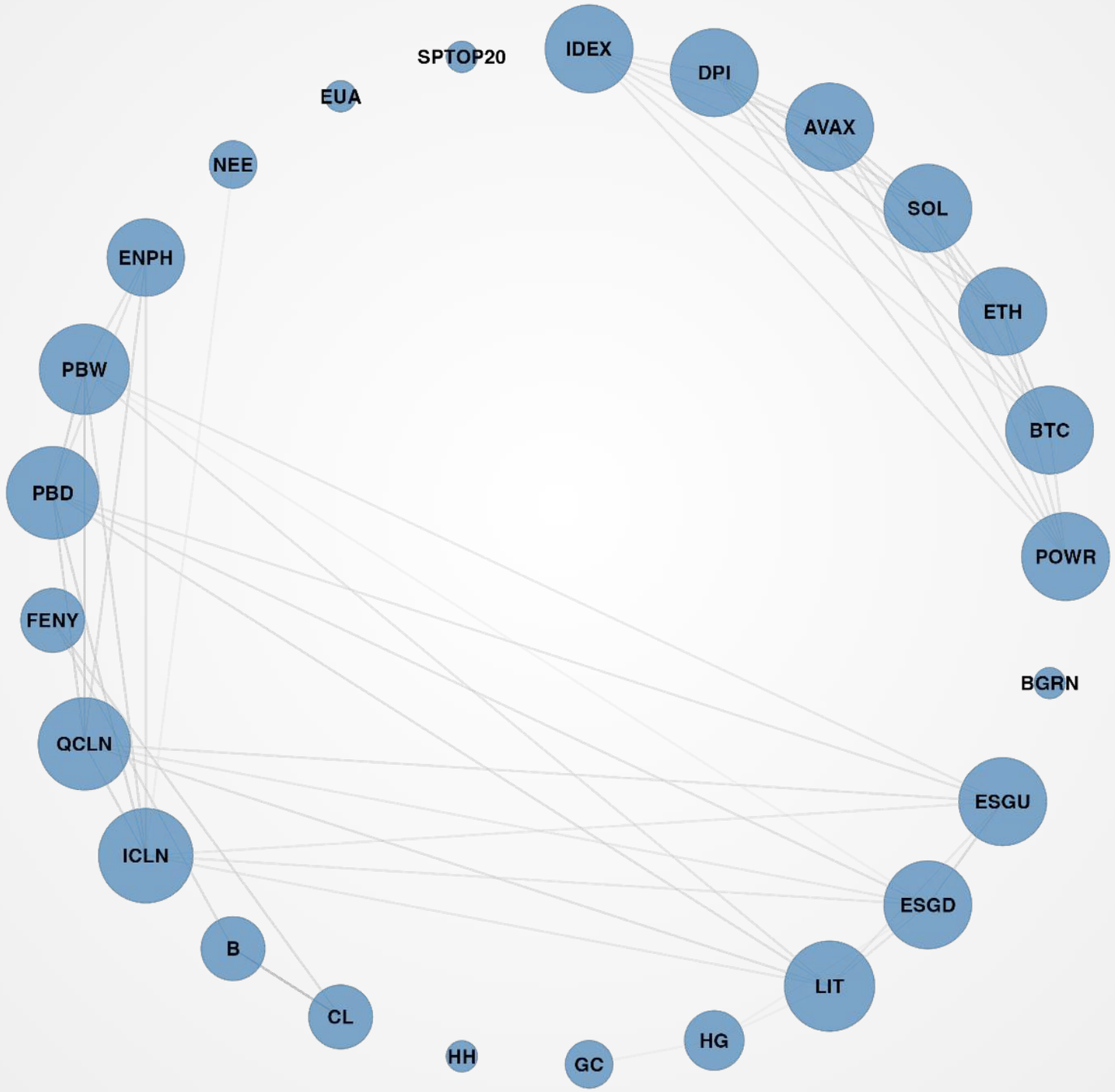
Power Ledger - Directional NET - Heatmap



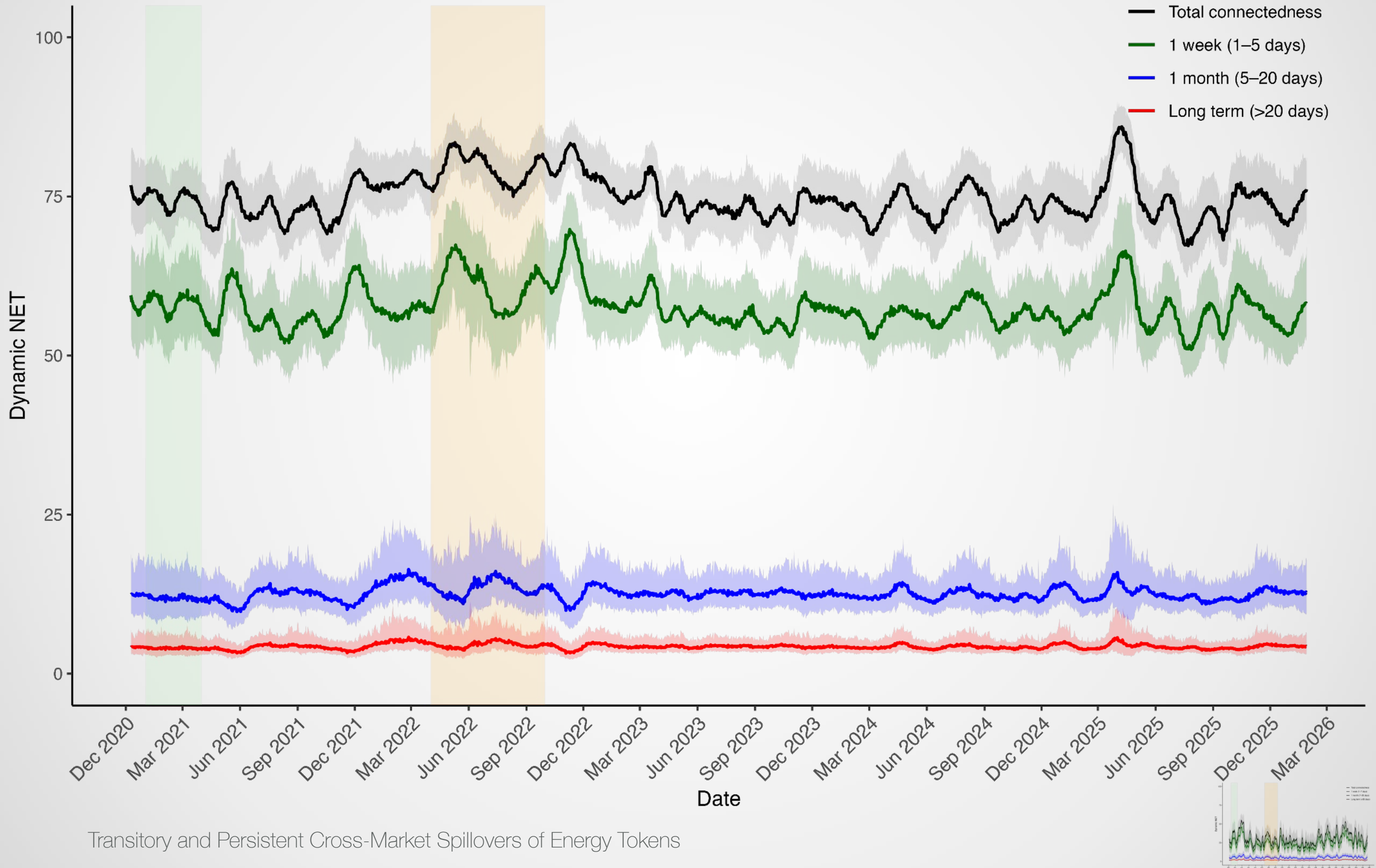
Transitory and Persistent Cross-Market Spillovers of Energy Tokens



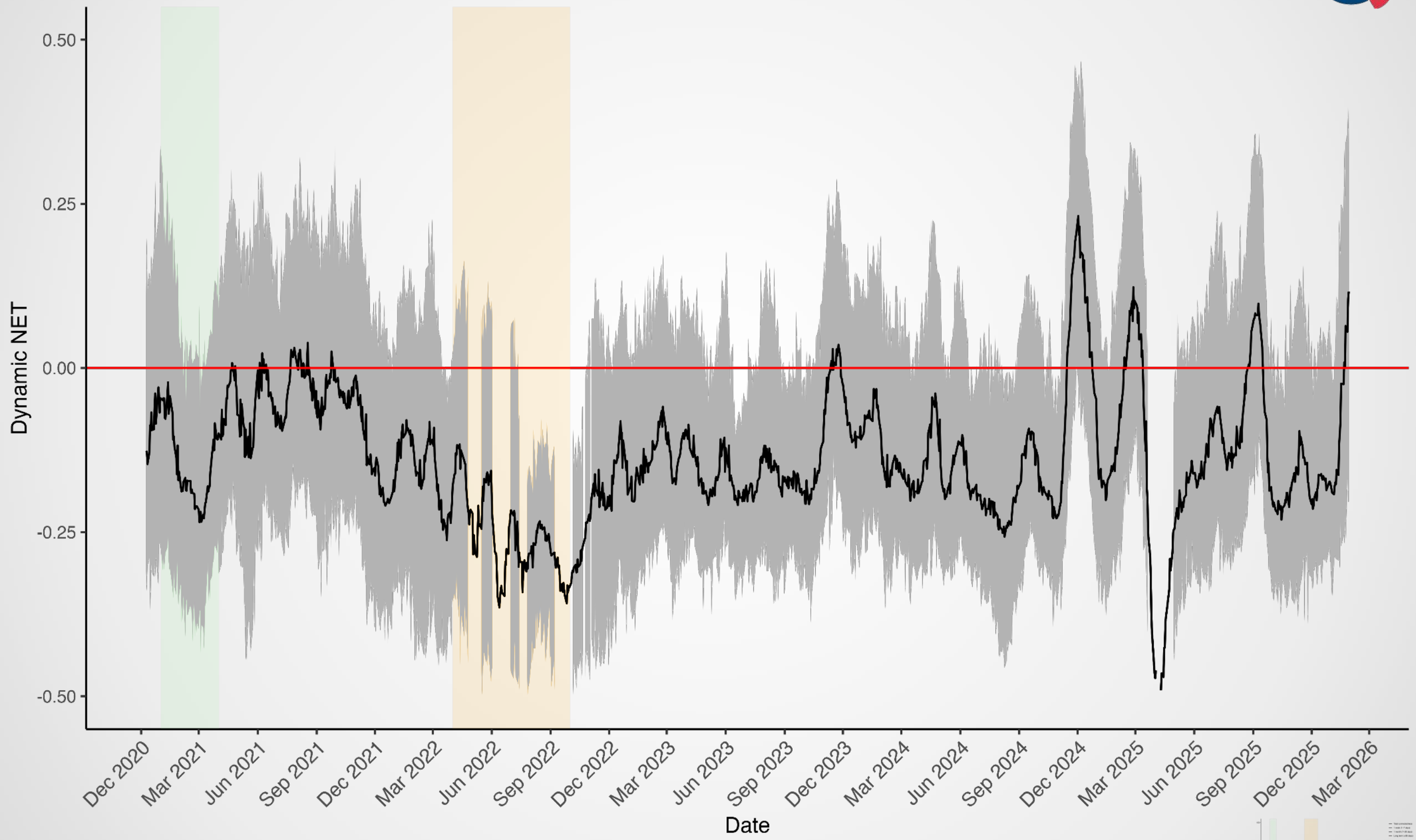
Power Ledger - Directional NET - Network



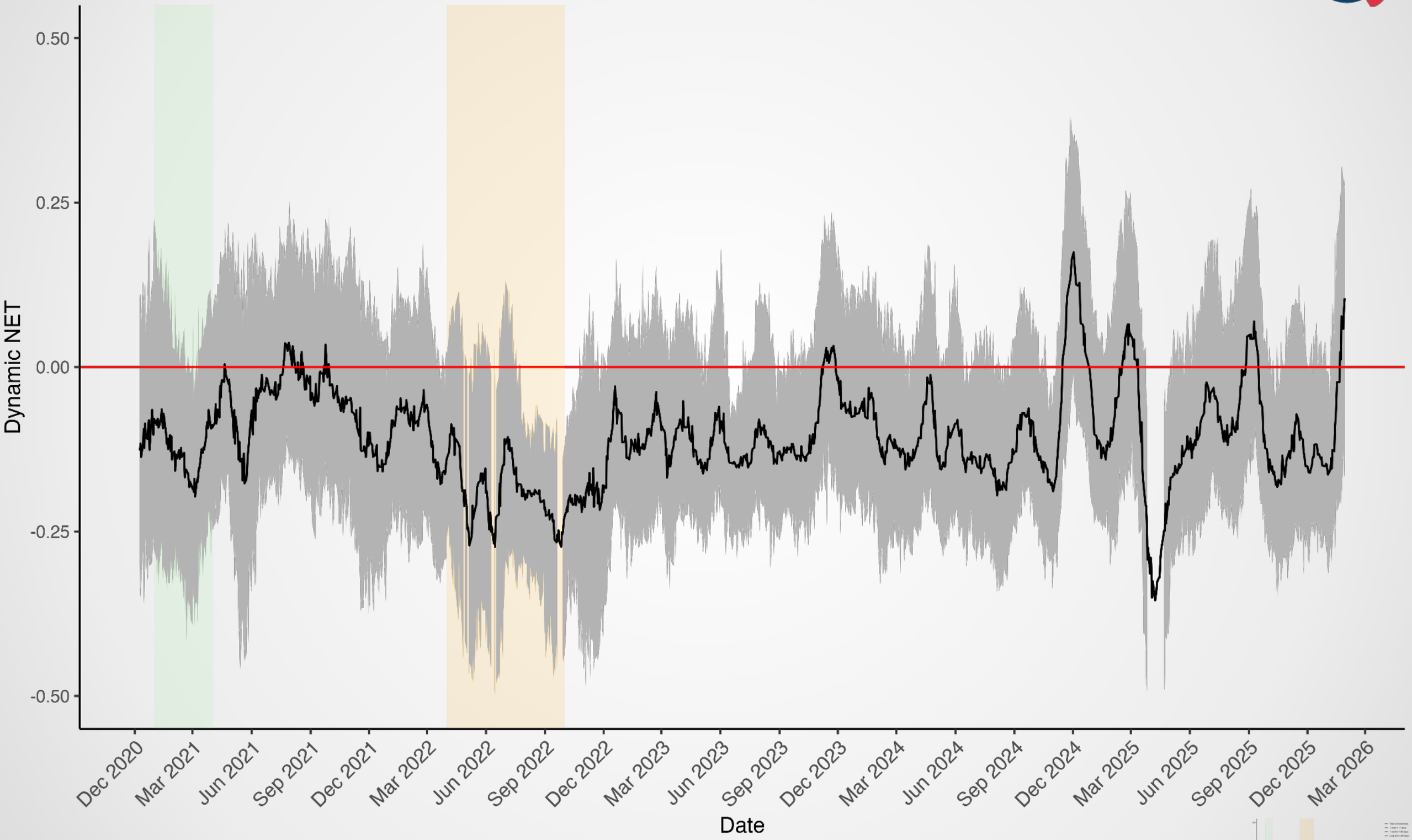
Energy Web Token - Dynamic Network Connectedness



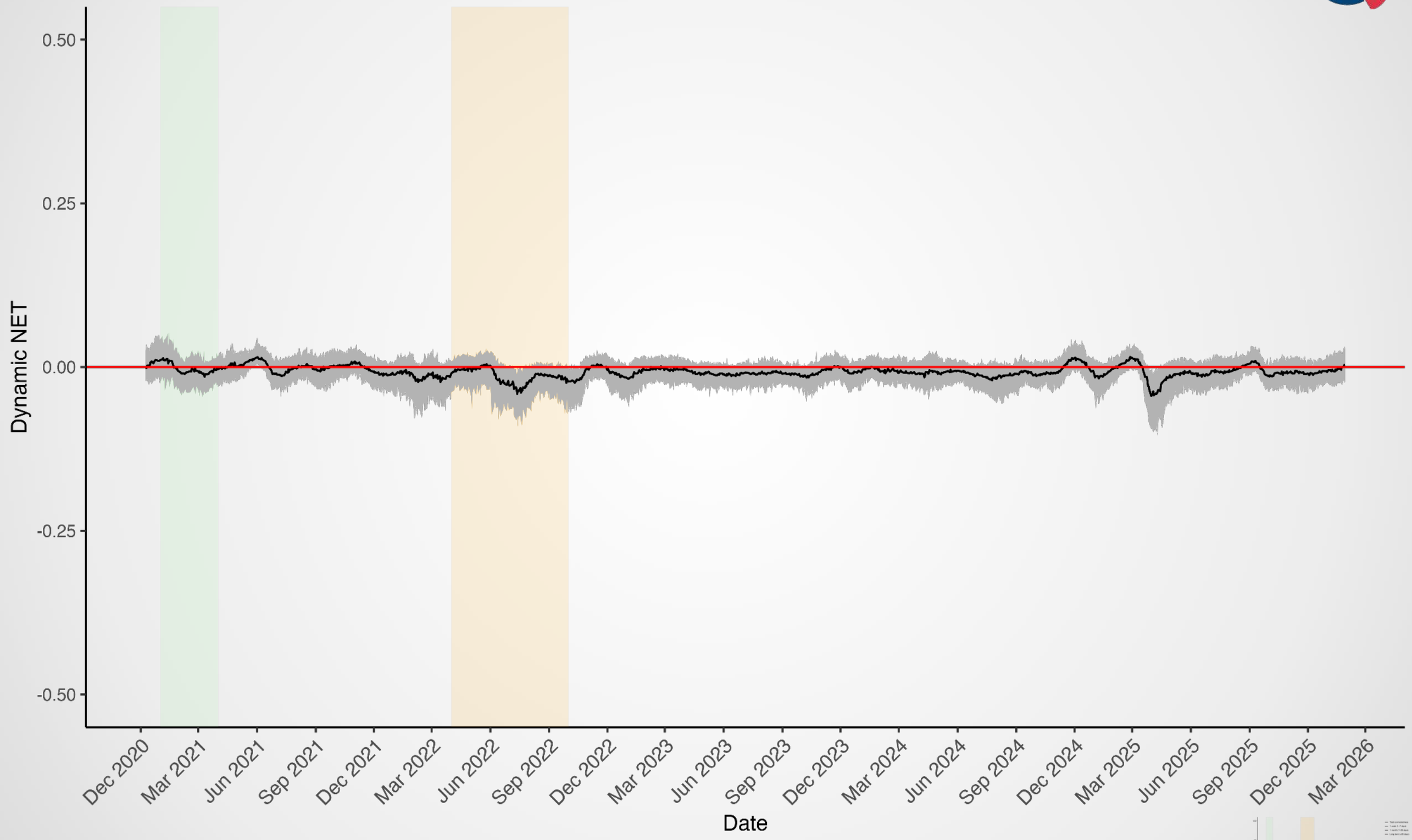
Energy Web Token- Directional NET - 1 week (1–5 days)



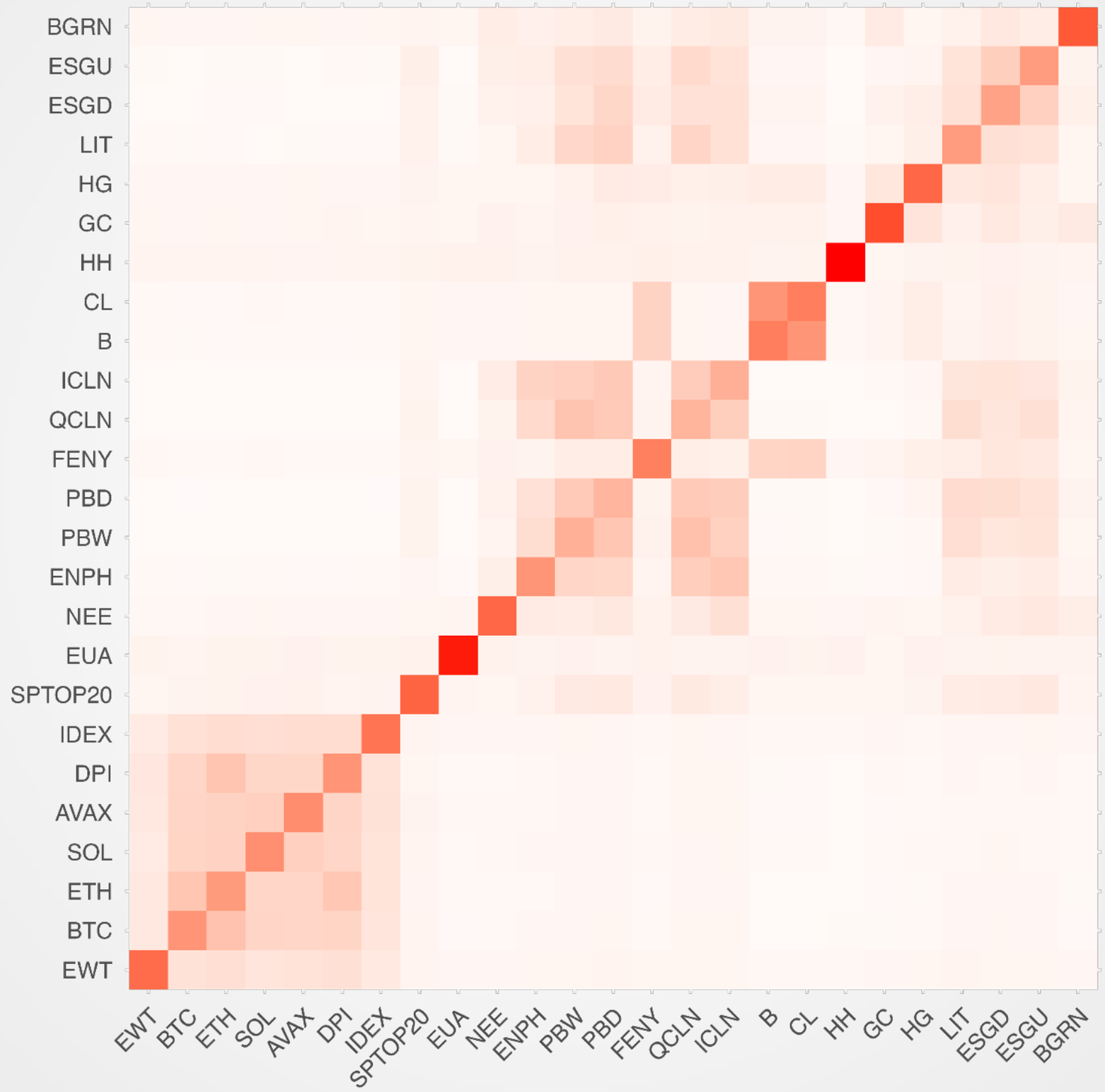
Energy Web Token - Directional NET - 1 month (5–20 days)



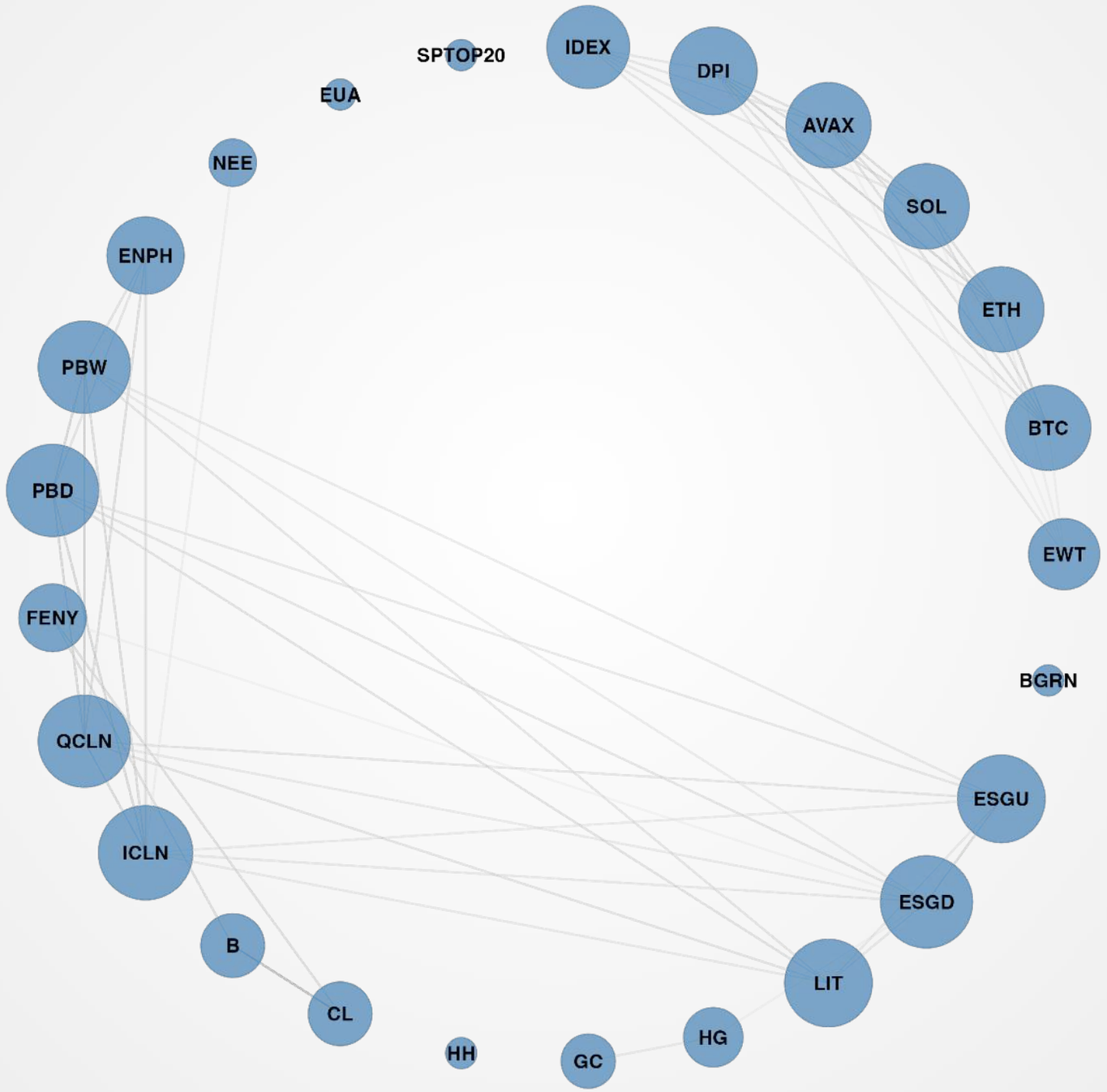
Energy Web Token - Directional NET - Long term (>20 days)



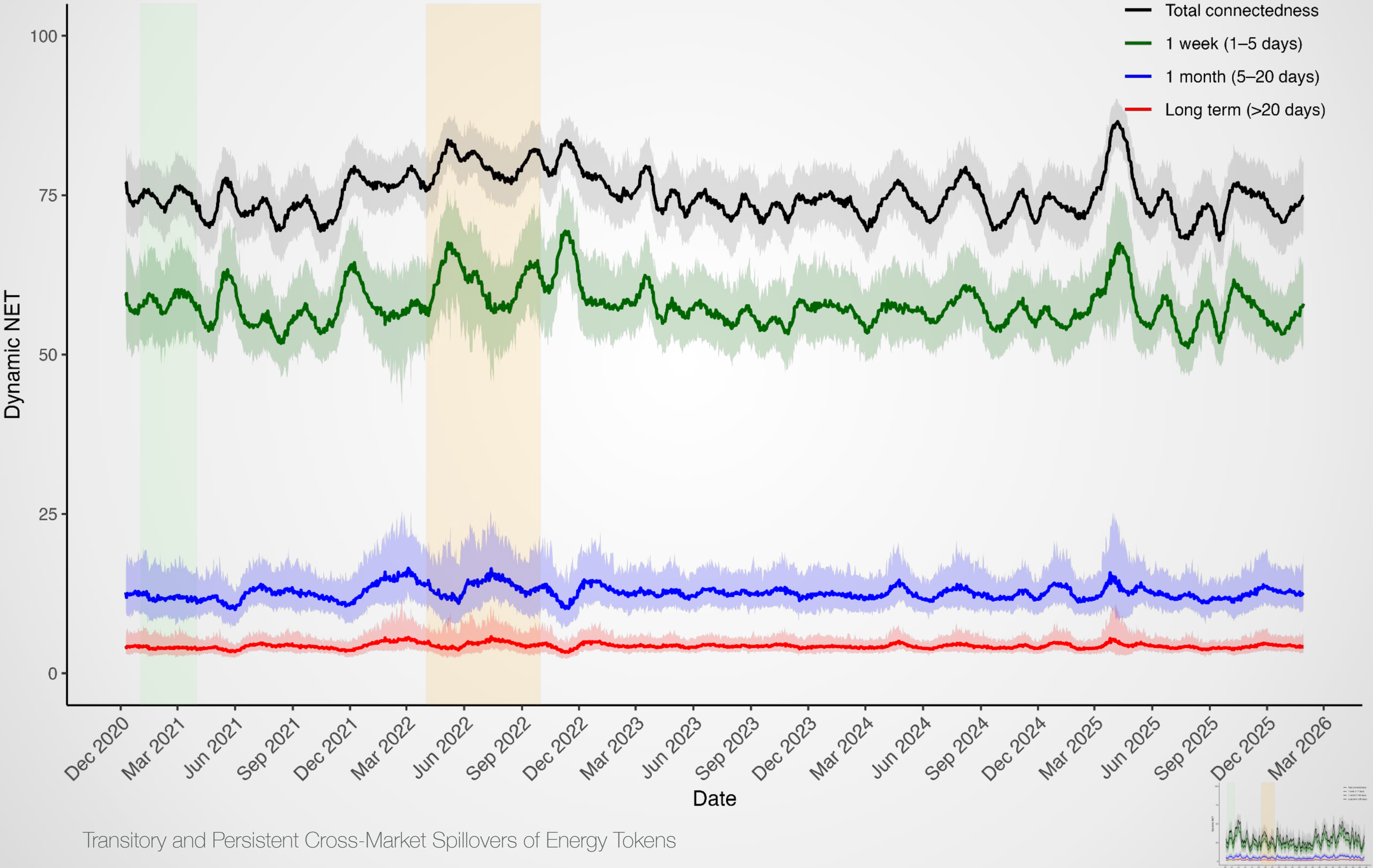
Energy Web Token - Directional NET - Heatmap



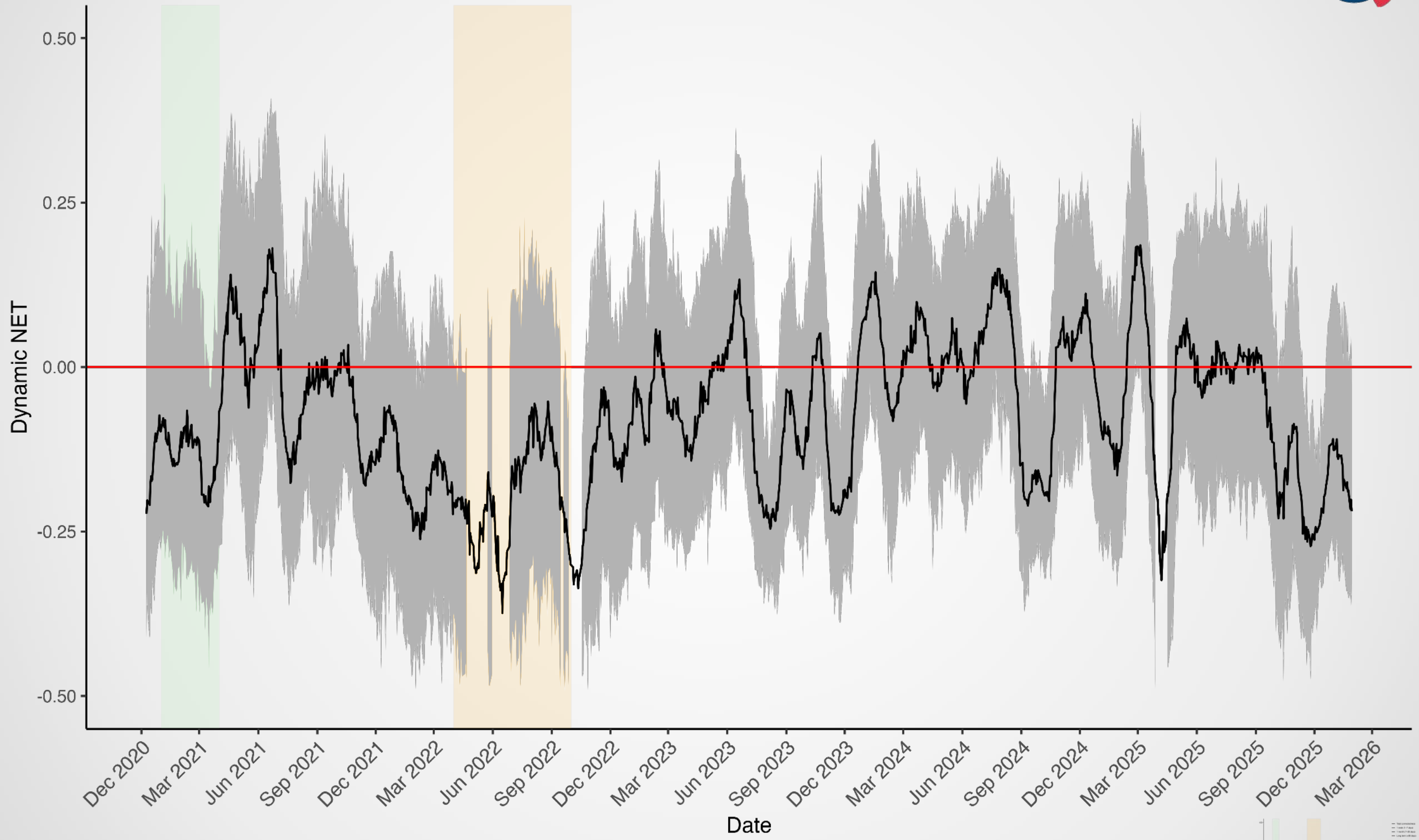
Energy Web Token - Directional NET - Network



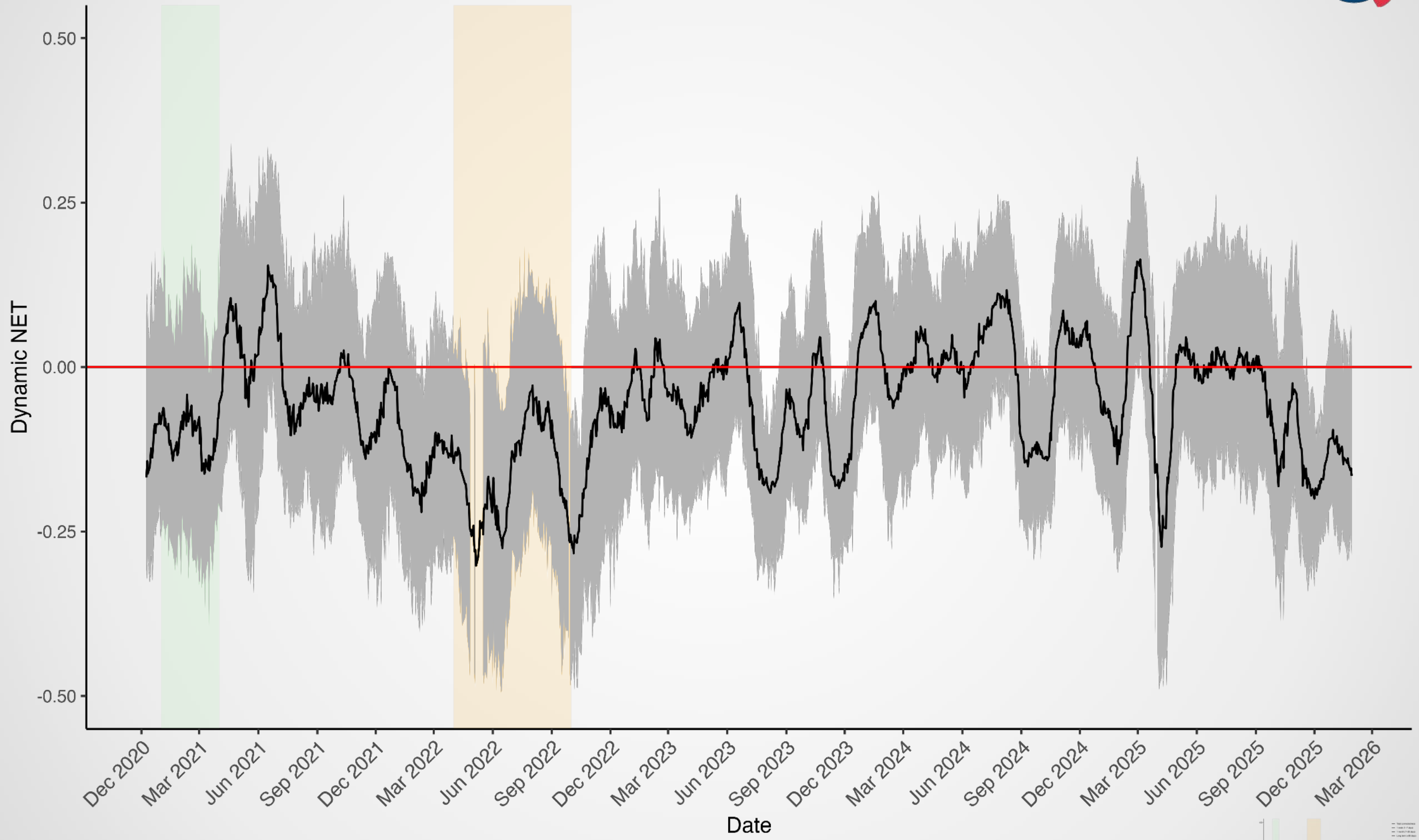
SunContract - Dynamic Network Connectedness



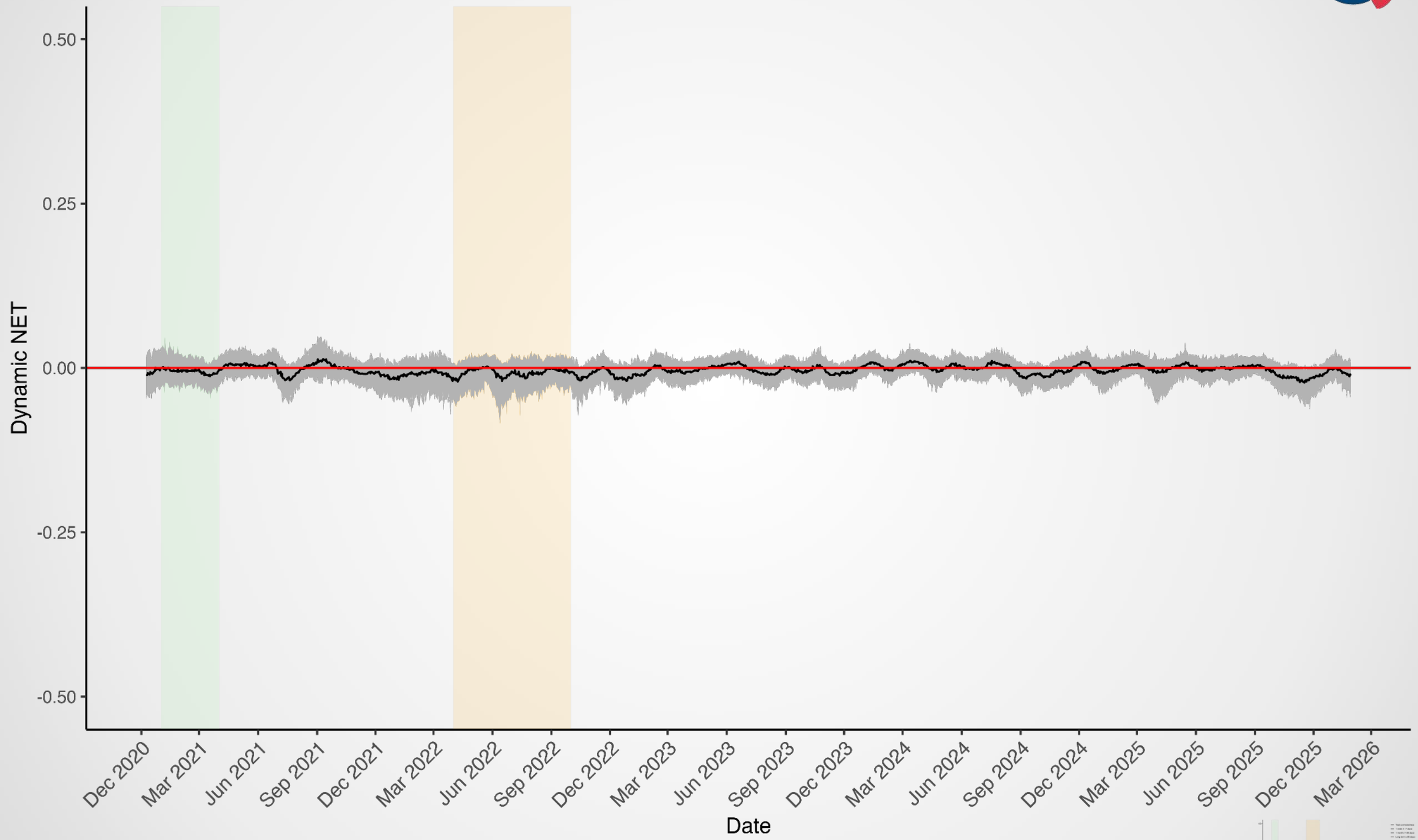
SunContract - Directional NET - 1 week (1–5 days)



SunContract - Directional NET - 1 month (5–20 days)



SunContract - Directional NET - Long term (>28 days)



Portfolio Selection Models

- Equal-Weight (EW) portfolio

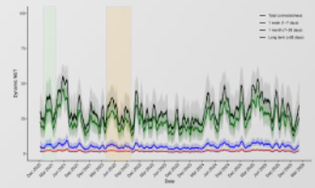
$$\mathbf{w}_t^{\mathrm{EW}} = \frac{1}{N}$$

- Global minimum-variance (GMV) portfolio,

$$\mathbf{w}_t^{\mathrm{GMV}} = \arg\min_{\mathbf{w}} \mathbf{w}'\hat{\Sigma}_t\mathbf{w}$$

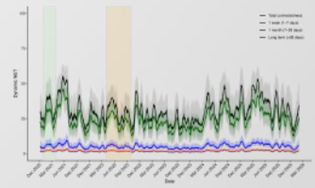
- Mean-variance-efficient (MVE) portfolio

$$\mathbf{w}_t^{\mathrm{MVE}} = \arg\max_{\mathbf{w}} \frac{\mathbf{w}'\hat{\mu}_t}{\sqrt{\mathbf{w}'\hat{\Sigma}_t\mathbf{w}}}$$



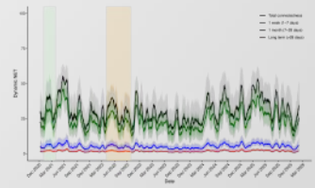
Performance Measures

- Mean return
- Volatility
- Maximum drawdown
- Value-at-Risk
- Expected Shortfall
- Sharpe Ratio
- Sortino ratio
- Calmar ratio



Full-sample portfolio performance

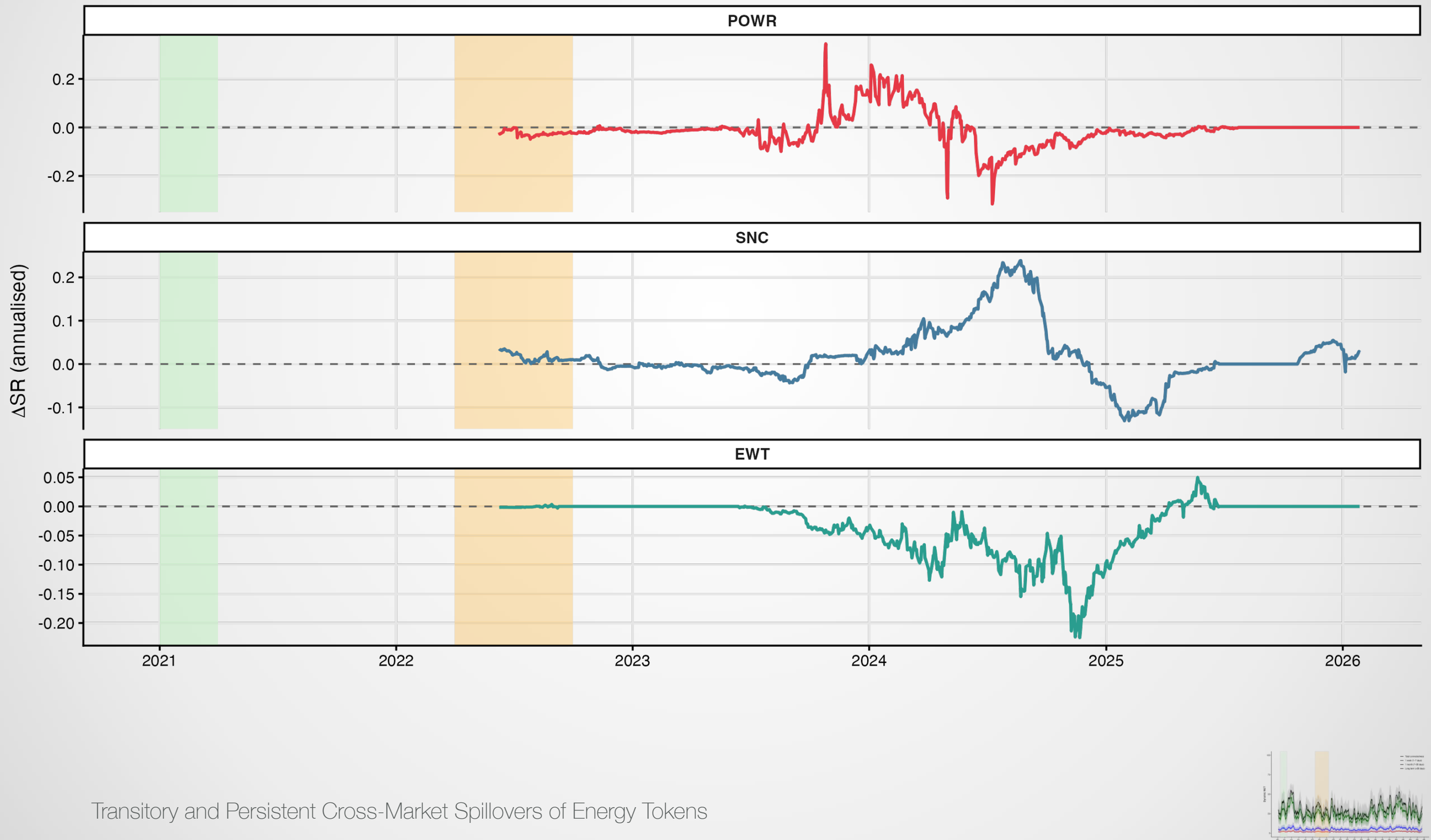
Token	Strategy	$\bar{r}_p$ (%)	σ_p (%)	SR	Sortino	MDD (%)	Calmar	$\text{VaR}_{5\%}$	$\text{CVaR}_{5\%}$	ΔSR
Baseline	EW	−8.01	24.50	−0.327	−0.447	50.65	−0.158	2.345	3.774	–
	GMV	0.71	5.64	0.127	0.187	18.05	0.040	0.590	0.805	–
	MVE	9.19	46.43	0.198	0.264	60.08	0.153	4.234	6.806	–
+POWR	EW	−9.38	25.85	−0.363	−0.491	52.60	−0.178	2.450	4.055	−0.036
	GMV	0.70	5.64	0.124	0.183	18.07	0.039	0.587	0.807	−0.003
	MVE	9.12	46.44	0.196	0.262	60.08	0.152	4.234	6.807	−0.002
+SNC	EW	−7.68	25.20	−0.305	−0.418	49.52	−0.155	2.447	3.894	0.022
	GMV	0.81	5.64	0.143	0.211	17.96	0.045	0.594	0.807	0.016
	MVE	10.34	46.34	0.223	0.295	58.09	0.178	4.192	6.803	0.025
+EWT	EW	−10.48	25.43	−0.412	−0.562	55.31	−0.189	2.512	3.956	−0.085
	GMV	0.60	5.65	0.106	0.156	18.06	0.033	0.594	0.806	−0.021
	MVE	9.19	46.43	0.198	0.264	60.08	0.153	4.234	6.806	≈0



Cumulative portfolio returns (GMV strategy)

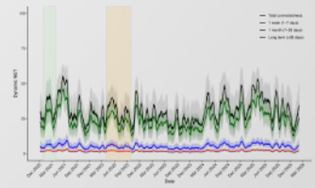


Rolling Sharpe ratio gain from adding each energy token



Conclusion

- Spillovers are strongly time-varying and **mainly short-run**
  - The transitory component dominates total connectedness
  - Persistent connectedness remains substantially lower and relatively stable
- **POWR and SNC** are the most systemically central transmitters in short-run windows
  - EWT becomes relevant mainly in stress episodes
- Connectedness is dominated by **transitory shocks**
  - Short-run band ≈ 60% of total, long-run ≈ 5%
  - Crises amplify short-run shocks, the long-run structure stays flat
- Diversification value is **token-specific**
  - **SNC:** only token with consistent positive ΔSR across **all data splits**
  - POWR: neutral
  - EWT: persistent net receiver, drags long-run Sharpe





Thank you for your attention

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